

SOLIDWORKS API (.NET VBA C#) — Reference (118/337): IFeatureManager (part 5 of 9)

SOLIDWORKS API Documentation (converted from CHM)

IFeatureManager (part 5 of 9) (Type)

SOLIDWORKS API Help

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InsertCoordinateSystem Method (IFeatureManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertCoordinateSystem Method (IFeatureManager)

Inserts a coordinate system feature.

.NET Syntax

Visual Basic (Declaration)

```
Function InsertCoordinateSystem( _  
    ByVal XFlippedIn As System.Boolean, _  
    ByVal YFlippedIn As System.Boolean, _  
    ByVal ZFlippedIn As System.Boolean _  
) As Feature
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager  
Dim XFlippedIn As System.Boolean  
Dim YFlippedIn As System.Boolean  
Dim ZFlippedIn As System.Boolean  
Dim value As Feature  
  
value = instance.InsertCoordinateSystem(XFlippedIn, YFlippedIn, ZFlippedIn)
```

C#

```
Feature InsertCoordinateSystem(  
    System.bool XFlippedIn,  
    System.bool YFlippedIn,  
    System.bool ZFlippedIn  
)
```

C++/CLI

```
Feature^ InsertCoordinateSystem(  
& System.bool XFlippedIn,  
& System.bool YFlippedIn,  
& System.bool ZFlippedIn  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

XFlippedIn
True to flip the x axis direction, false to not
YFlippedIn
True to flip the y axis direction, false to not
ZFlippedIn
True to flip the z axis direction, false to not

Return Value

[Feature](#)

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::InsertCoordinateSystem](#).

Example

[Insert Coordinate System Feature at Center of Mass \(VBA\)](#).

Remarks

Make the selections using [IModelDocExtension::SelectByID2](#) with a mark of:

- 1 - Origin
- 2 - X axis
- 4 - Y axis
- 8 - Z axis

This method:

- does not require all three axis to be selected. The behavior is the same as interactively creating a coordinate system. See the SOLIDWORKS Help for more information.
- works in section-view mode, but not if temporary geometry that only exists in section-view mode is selected.

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[ICoordinateSystemFeatureData Interface](#)
[IFeatureManager::CreateCoordinateSystem Method](#)

Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertCosmeticThread Method (IFeatureManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertCosmeticThread Method (IFeatureManager)

Obsolete. Superseded by [IFeatureManager::InsertCosmeticThread2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function InsertCosmeticThread( _  
    ByVal Type As System.Short, _  
    ByVal Depth As System.Double, _  
    ByVal Length As System.Double, _  
    ByVal Note As System.String _  
) As System.String
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager  
Dim Type As System.Short  
Dim Depth As System.Double  
Dim Length As System.Double  
Dim Note As System.String  
Dim value As System.String
```

```
value = instance.InsertCosmeticThread(Type, Depth, Length, Note)
```

C#

```
System.string InsertCosmeticThread(  
    System.short Type,  
    System.double Depth,  
    System.double Length,  
    System.string Note  
)
```

C++/CLI

```
System.String^ InsertCosmeticThread(  
& System.short Type,  
& System.double Depth,  
& System.double Length,  
& System.String^ Note  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Type

Depth

Length

Note

Visual Basic for Applications (VBA) Syntax

See [FeatureManager.InsertCosmeticThread](#).

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)

Availability

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertCosmeticThread2 Method (IFeatureManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertCosmeticThread2 Method (IFeatureManager)

Obsolete. Superseded by [IFeatureManager.InsertCosmeticThread3](#).

.NET Syntax

Visual Basic (Declaration)

```
Function InsertCosmeticThread2( _  
    ByVal Type As System.Short, _  
    ByVal Depth As System.Double, _  
    ByVal Length As System.Double, _  
    ByVal Note As System.String _  
) As Feature
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager  
Dim Type As System.Short  
Dim Depth As System.Double  
Dim Length As System.Double  
Dim Note As System.String  
Dim value As Feature
```

```
value = instance.InsertCosmeticThread2(Type, Depth, Length, Note)
```

C#

```
Feature InsertCosmeticThread2(  
    System.short Type,  
    System.double Depth,  
    System.double Length,  
    System.string Note  
)
```

C++/CLI

```
Feature^ InsertCosmeticThread2(  
& System.short Type,  
& System.double Depth,  
& System.double Length,
```

```
& System.String^ Note  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Type

Type as defined in [swCosmeticThreadType](#) [e](#)

Depth

Diameter of the cosmetic thread

Length

Length of the cosmetic thread

Note

Callout text to display in the drawing document

Return Value

Pointer to the [IFeature](#) object

Visual Basic for Applications (VBA) Syntax

See [FeatureManager.InsertCosmeticThread2](#).

See Also

[IFeatureManager Interface](#)

[IFeatureManager Members](#)

[ICosmeticThreadFeatureData Interface](#)

Availability

SOLIDWORKS 2005 FCS, Revision Number 13.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertCosmeticThread3 Method (IFeatureManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertCosmeticThread3 Method (IFeatureManager)

Inserts a cosmetic thread.

.NET Syntax

Visual Basic (Declaration)

```
Function InsertCosmeticThread3( _  
    ByVal Standard As System.Integer, _  
    ByVal StandardType As System.String, _  
    ByVal Size As System.String, _  
    ByVal Diameter As System.Double, _  
    ByVal EndType As System.Integer, _  
    ByVal Depth As System.Double, _  
    ByVal Note As System.String _  
) As Feature
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager  
Dim Standard As System.Integer  
Dim StandardType As System.String  
Dim Size As System.String  
Dim Diameter As System.Double  
Dim EndType As System.Integer  
Dim Depth As System.Double  
Dim Note As System.String  
Dim value As Feature
```

```
value = instance.InsertCosmeticThread3(Standard, StandardType, Size, Diameter, EndType, Depth,  
Note)
```

C#

```
Feature InsertCosmeticThread3(
    System.int Standard,
    System.string StandardType,
    System.string Size,
    System.double Diameter,
    System.int EndType,
    System.double Depth,
    System.string Note
)
```

C++/CLI

```
Feature^ InsertCosmeticThread3(
    & System.int Standard,
    & System.String^ StandardType,
    & System.String^ Size,
    & System.double Diameter,
    & System.int EndType,
    & System.double Depth,
    & System.String^ Note
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Standard

Thread standard as defined by [swCosmeticStandardType_e](#)

StandardType

Thread type for Standard

Size

Thread size for the specified Standard

Diameter

Thread diameter

EndType

End condition as defined by [swCosmeticEndConditions_e](#)

Depth

Depth of the cosmetic thread; valid only for EndType = swCosmeticEndConditions_e.swEndConditionBlind or swCosmeticEndConditions_e.swEndConditionBlindUptoNext or swCosmeticEndConditions_e.swEndConditionBlind2Dia

Note

Callout text to display in the drawing document

Return Value

[IFeature](#)

Visual Basic for Applications (VBA) Syntax

See [FeatureManager.InsertCosmeticThread3](#).

Example

[Traverse All Cosmetic Threads \(VBA\)](#)

[Traverse All Cosmetic Threads \(VB.NET\)](#)

[Traverse All Cosmetic Threads \(C#\)](#)

See Also

[IFeatureManager Interface](#)

[IFeatureManager Members](#)

[ICosmeticThreadFeatureData Interface](#)

Availability

SOLIDWORKS 2015 FCS, Revision Number 23.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertCosmeticWeldBead Method (IFeatureManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertCosmeticWeldBead Method (IFeatureManager)

Obsolete. Superseded by [IFeatureManager.InsertCosmeticWeldBead2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function InsertCosmeticWeldBead( _  
    ByVal WeldSize As System.Double _  
) As Feature
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager  
Dim WeldSize As System.Double  
Dim value As Feature  
  
value = instance.InsertCosmeticWeldBead(WeldSize)
```

C#

```
Feature InsertCosmeticWeldBead(  
    System.double WeldSize  
)
```

C++/CLI

```
Feature^ InsertCosmeticWeldBead(  
& System.double WeldSize  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

WeldSize
Size of the weld bead

Return Value

[IFeature](#)

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::InsertCosmeticWeldBead](#).

Example

[Insert Cosmetic Weld Bead \(VBA\)](#)
[Insert Cosmetic Weld Bead \(VB.NET\)](#)
[Insert Cosmetic Weld Bead \(C#\)](#)

Remarks

Before calling this method, select the contact faces for the weld bead.

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)

Availability

SOLIDWORKS 2013 FCS, Revision Number 21.0

SOLIDWORKS API Help

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InsertCosmeticWeldBead2 Method (IFeatureManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertCosmeticWeldBead2 Method (IFeatureManager)

Inserts a cosmetic weld bead using either weld geometry or a weld path.

.NET Syntax

Visual Basic (Declaration)

```
Function InsertCosmeticWeldBead2( _  
    ByVal WeldMode As System.Integer, _  
    ByVal WeldFromFaceOrEdgeSet As System.Object, _  
    ByVal WeldToFaceOrEdgeSet As System.Object, _  
    ByVal WeldSize As System.Double _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager  
Dim WeldMode As System.Integer  
Dim WeldFromFaceOrEdgeSet As System.Object  
Dim WeldToFaceOrEdgeSet As System.Object  
Dim WeldSize As System.Double  
Dim value As System.Object  
  
value = instance.InsertCosmeticWeldBead2(WeldMode, WeldFromFaceOrEdgeSet, WeldToFaceOrEdgeSet, WeldSize)
```

C#

```
System.object InsertCosmeticWeldBead2(  
    System.int WeldMode,  
    System.object WeldFromFaceOrEdgeSet,  
    System.object WeldToFaceOrEdgeSet,  
    System.double WeldSize  
)
```

C++/CLI

```
System.Object^ InsertCosmeticWeldBead2(  
& System.int WeldMode,  
& System.Object^ WeldFromFaceOrEdgeSet,  
& System.Object^ WeldToFaceOrEdgeSet,  
& System.double WeldSize  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

WeldMode
Weld mode as defined in [swCosmeticWeldBeadMode](#) e
WeldFromFaceOrEdgeSet
Array of weld-from entities (see **Remarks**)
WeldToFaceOrEdgeSet
Array of weld-to entities or Nothing or null (see **Remarks**)

WeldSize
Size of weld bead

Return Value

Array of weld bead [features](#)

 Visual Basic for Applications (VBA) Syntax

See [FeatureManager.InsertCosmeticWeldBead2](#).

 Example

[Insert Cosmetic Weld Bead Using Geometric Entities \(C#\)](#)
[Insert Cosmetic Weld Bead Using Geometric Entities \(VB.NET\)](#)
[Insert Cosmetic Weld Bead Using Geometric Entities \(VBA\)](#)

 Remarks

WeldMode	Selected entities	Selection marks
swCosmeticWeldBeadMode_e_swCosmeticWeldBeadMode_WeldGeometry	Either faces or edges as weld-from and weld-to entities NOTE: Both weld-to and weld-from entities must all be the same type of entities; e.g., all faces or all edges.	• 4 for each weld-from entity in WeldFromFaceOrEdgeSe • 8 for each weld-to entity if WeldToFaceOrEdgeSet

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)

Availability

SOLIDWORKS 2015 FCS, Revision Number 23.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertCrossBreak Method (IFeatureManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertCrossBreak Method (IFeatureManager)

Inserts a cross break feature on the selected face in a sheet metal part.

.NET Syntax

Visual Basic (Declaration)

```
Function InsertCrossBreak( _
    ByVal Angle As System.Double, _
    ByVal Radius As System.Double _
) As Feature
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager
Dim Angle As System.Double
Dim Radius As System.Double
Dim value As Feature
```

```
value = instance.InsertCrossBreak(Angle, Radius)
```

C#

```
Feature InsertCrossBreak(
    System.double Angle,
    System.double Radius
)
```

C++/CLI

```
Feature^ InsertCrossBreak(
    & System.double Angle,
    & System.double Radius
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Angle
 Break angle
Radius
 Break radius

Return Value

Cross break [feature](#)

Visual Basic for Applications (VBA) Syntax

See [FeatureManager.InsertCrossBreak](#).

Example

[Get Cross Break Feature Data in Sheet Metal Part \(C#\)](#)
[Get Cross Break Feature Data in Sheet Metal Part \(VB.NET\)](#)
[Get Cross Break Feature Data in Sheet Metal Part \(VBA\)](#)

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[ICrossBreakFeatureData Interface](#)

Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

InsertCutBlend Method (IFeatureManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertCutBlend Method (IFeatureManager)

Inserts a lofted cut based on the selected profiles, centerline, and guide curves.

.NET Syntax

Visual Basic (Declaration)

```
Function InsertCutBlend( _  
    ByVal Closed As System.Boolean, _  
    ByVal KeepTangency As System.Boolean, _  
    ByVal ForceNonRational As System.Boolean, _  
    ByVal TessToleranceFactor As System.Double, _  
    ByVal StartMatchingType As System.Short, _  
    ByVal EndMatchingType As System.Short, _  
    ByVal IsThinBody As System.Boolean, _  
    ByVal Thickness1 As System.Double, _  
    ByVal Thickness2 As System.Double, _  
    ByVal ThinType As System.Short, _  
    ByVal UseFeatScope As System.Boolean, _  
    ByVal UseAutoSelect As System.Boolean _  
) As Feature
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager  
Dim Closed As System.Boolean  
Dim KeepTangency As System.Boolean  
Dim ForceNonRational As System.Boolean  
Dim TessToleranceFactor As System.Double  
Dim StartMatchingType As System.Short  
Dim EndMatchingType As System.Short  
Dim IsThinBody As System.Boolean  
Dim Thickness1 As System.Double  
Dim Thickness2 As System.Double  
Dim ThinType As System.Short  
Dim UseFeatScope As System.Boolean  
Dim UseAutoSelect As System.Boolean  
Dim value As Feature
```

```
value = instance.InsertCutBlend(Closed, KeepTangency, ForceNonRational, TessToleranceFactor,  
StartMatchingType, EndMatchingType, IsThinBody, Thickness1, Thickness2, ThinType, UseFeatScope,  
UseAutoSelect)
```

C#

```
Feature InsertCutBlend(  
    System.bool Closed,  
    System.bool KeepTangency,  
    System.bool ForceNonRational,  
    System.double TessToleranceFactor,  
    System.short StartMatchingType,
```

```

    System.short EndMatchingType,
    System.bool IsThinBody,
    System.double Thickness1,
    System.double Thickness2,
    System.short ThinType,
    System.bool UseFeatScope,
    System.bool UseAutoSelect
)

```

C++/CLI

```

Feature^ InsertCutBlend(
& System.bool Closed,
& System.bool KeepTangency,
& System.bool ForceNonRational,
& System.double TessToleranceFactor,
& System.short StartMatchingType,
& System.short EndMatchingType,
& System.bool IsThinBody,
& System.double Thickness1,
& System.double Thickness2,
& System.short ThinType,
& System.bool UseFeatScope,
& System.bool UseAutoSelect
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Closed

True if you want the loft to be closed, false to leave it open (see **Remarks**)

KeepTangency

Controls whether the section curves are tangent (see Remarks)

ForceNonRational

True to force the resulting surface to be non-rational; false to not

TessToleranceFactor

A factor to control the number of intermediate sections used for loft with centerline; the default value is 1.0; the greater the variable, the more intermediate sections are created

StartMatchingType

Tangency type at the start profile (see Remarks)

EndMatchingType

Tangency type at the end profile (see Remarks)

IsThinBody

True if this feature is a thin body, false is not

Thickness1

Thickness value for the first direction

Thickness2

Thickness value for the second direction

ThinType

Thin wall type:

- 0 = One direction
- 1 = One direction reverse
- 2 = Mid-plane
- 3 = Two direction

UseFeatScope

True if the feature only affects selected bodies, false if the feature affects all bodies

UseAutoSelect

True to automatically select all bodies and have the feature affect those bodies, false to select the bodies the feature affects (see Remarks)

Return Value

[Feature](#)

Visual Basic for Applications (VBA) Syntax

See [FeatureManager.InsertCutBlend](#).

Remarks

Selection of guide curves and centerline is optional; however, selection of the profiles must be in an order consistent with the desired direction of the loft.

Use of guide curves is strongly recommended, especially when selection of profiles is done in the FeatureManager design tree.

You can use any number of profiles; however, if you selected only one profile, then any selected guide curves must be closed curves.

If Closed is True and you selected less than three profiles, then any selected guide curves must be closed curves.

If the section curves are tangent, then KeepTangency controls whether the resulting surfaces are also be tangent. Specify True to maintain the tangency as seen in the section curves; false to not. When generating tangent surfaces, SOLIDWORKS maintains planar and cylindrical surface

shapes if the section curves exhibit these characteristics.

Use [IModelDocExtension::SelectById2](#) to select the profiles and guide curves. The mark for the profile selections should be 1; the mark for any guide curve selection, if provided, should be a 2; the mark for the centerline selection, if provided, should be a 4; the mark for the start tangency vector selection, if provided, should be a 8; the mark for the start tangency faces selection, if provided, should be a 16 (not available); the mark for the end tangency vector selection, if provided, should be a 32; the mark for the end tangency faces selection, if provided, should be a 64 (not available); linear edge, sketch line, axis, plane and planar faces are qualified for tangency vector sections.

The tangency type arguments can take the following values:

- 0 - none
- 1 - tangent to the normal of the profile
- 2 - tangent to a selected vector
- 3 - tangency to all the adjacent faces sharing an edge with the start profile
- 4 - tangent to some of the selected faces sharing an edge with the start profile (not currently available)

When UseAutoSelect is false, the user must select the bodies that the feature will affect.

When using cut or cavity features that result in multiple bodies, you cannot select to keep all of the resulting bodies or one or more selected bodies.

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[ILoftFeatureData Interface](#)
[IFeatureManager::InsertProtrusionBlend Method](#)

Availability

SOLIDWORKS 2003 FCS, Revision Number 11.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertCutSurface Method (IFeatureManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertCutSurface Method (IFeatureManager)

Inserts a surface-cut feature using the preselected surface or plane.

.NET Syntax

Visual Basic (Declaration)

```
Function InsertCutSurface( _  
    ByVal Flip As System.Boolean, _  
    ByVal KeepPieceIndex As System.Integer, _  
    ByVal UseFeatScope As System.Boolean, _  
    ByVal UseAutoSelect As System.Boolean, _  
    ByVal Bodies As System.Object, _  
    ByRef Error As System.Integer _  
) As Feature
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager  
Dim Flip As System.Boolean  
Dim KeepPieceIndex As System.Integer  
Dim UseFeatScope As System.Boolean  
Dim UseAutoSelect As System.Boolean  
Dim Bodies As System.Object  
Dim Error As System.Integer  
Dim value As Feature
```

```
value = instance.InsertCutSurface(Flip, KeepPieceIndex, UseFeatScope, UseAutoSelect, Bodies,  
Error)
```

C#

```
Feature InsertCutSurface(
    System.bool Flip,
    System.int KeepPieceIndex,
    System.bool UseFeatScope,
    System.bool UseAutoSelect,
    System.object Bodies,
    out System.int Error
)
```

C++/CLI

```
Feature^ InsertCutSurface(
    & System.bool Flip,
    & System.int KeepPieceIndex,
    & System.bool UseFeatScope,
    & System.bool UseAutoSelect,
    & System.Object^ Bodies,
    & [Out] System.int Error
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Flip

True to flip the direction of the cut, false to not

KeepPieceIndex

Piece to keep if there is ambiguity (see **Remarks**)

UseFeatScope

True to cut only the bodies passed to Bodies that intersect with the cut, false to cut all bodies that intersect with the cut

UseAutoSelect

True to automatically cut all bodies that intersect with the cut, false to cut only the bodies passed to Bodies that intersect with the cut

Bodies

- Array of specific [bodies](#) that intersect with the cut
 - UseFeatScope = true
 - UseAutoSelect = false

- OR -

- Empty array indicating to cut all bodies that intersect with the cut
 - UseFeatScope = false
 - UseAutoSelect = true

Error

Status of the cut as defined in [swSurfaceCutFeatureError_e](#)

Return Value

Surface-cut [feature](#)

Visual Basic for Applications (VBA) Syntax

See [FeatureManager.InsertCutSurface](#).

Example

[Insert Surface-cut Feature \(C#\)](#)
[Insert Surface-cut Feature \(VB.NET\)](#)
[Insert Surface-cut Feature \(VBA\)](#)

Remarks

When there is ambiguity in the result of a cut, KeepPieceIndex is used to resolve which of the possible results is used. This can be set to -1 if there is no ambiguity; otherwise, it should be the index of the result, starting from 0 (up to one less than the possible number of outcomes).

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[ISurfaceCutFeatureData Interface](#)

Availability

InsertCutSwept Method (IFeatureManager)

[See Also](#)[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertCutSwept Method (IFeatureManager)**Obsolete.** Superseded by [IFeatureManager.InsertCutSwept3](#).

.NET Syntax

Visual Basic (Declaration)

```

Function InsertCutSwept( _
    ByVal Propagate As System.Boolean, _
    ByVal Alignment As System.Boolean, _
    ByVal TwistCtrlOption As System.Short, _
    ByVal KeepTangency As System.Boolean, _
    ByVal ForceNonRational As System.Boolean, _
    ByVal StartMatchingType As System.Short, _
    ByVal EndMatchingType As System.Short, _
    ByVal IsThinBody As System.Boolean, _
    ByVal Thickness1 As System.Double, _
    ByVal Thickness2 As System.Double, _
    ByVal ThinType As System.Short, _
    ByVal UseFeatScope As System.Boolean, _
    ByVal UseAutoSelect As System.Boolean _
) As Feature

```

Visual Basic (Usage)



```

Dim instance As IFeatureManager
Dim Propagate As System.Boolean
Dim Alignment As System.Boolean
Dim TwistCtrlOption As System.Short
Dim KeepTangency As System.Boolean
Dim ForceNonRational As System.Boolean
Dim StartMatchingType As System.Short
Dim EndMatchingType As System.Short
Dim IsThinBody As System.Boolean
Dim Thickness1 As System.Double
Dim Thickness2 As System.Double
Dim ThinType As System.Short
Dim UseFeatScope As System.Boolean
Dim UseAutoSelect As System.Boolean
Dim value As Feature

value = instance.InsertCutSwept(Propagate, Alignment, TwistCtrlOption, KeepTangency,
    ForceNonRational, StartMatchingType, EndMatchingType, IsThinBody, Thickness1, Thickness2,
    ThinType, UseFeatScope, UseAutoSelect)

```

C#

```

Feature InsertCutSwept(
    System.bool Propagate,
    System.bool Alignment,
    System.short TwistCtrlOption,
    System.bool KeepTangency,
    System.bool ForceNonRational,
    System.short StartMatchingType,
    System.short EndMatchingType,
    System.bool IsThinBody,
    System.double Thickness1,
    System.double Thickness2,
    System.short ThinType,
    System.bool UseFeatScope,
    System.bool UseAutoSelect
)

```

C++/CLI

```

Feature^ InsertCutSwept(
& System.bool Propagate,
& System.bool Alignment,
& System.short TwistCtrlOption,
& System.bool KeepTangency,
& System.bool ForceNonRational,
& System.short StartMatchingType,
& System.short EndMatchingType,
& System.bool IsThinBody,
& System.double Thickness1,
& System.double Thickness2,
& System.short ThinType,
& System.bool UseFeatScope,
& System.bool UseAutoSelect
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Propagate

Alignment

TwistCtrlOption

KeepTangency

ForceNonRational

StartMatchingType

EndMatchingType

IsThinBody

Thickness1

Thickness2

ThinType

UseFeatScope

UseAutoSelect

Visual Basic for Applications (VBA) Syntax

See [FeatureManager.InsertCutSwept](#).

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertCutSwept2 Method (IFeatureManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertCutSwept2 Method (IFeatureManager)

Obsolete. Superseded by [IFeatureManager.InsertCutSwept3](#).

.NET Syntax

Visual Basic (Declaration)

```

Function InsertCutSwept2( _
    ByVal Propagate As System.Boolean, _
    ByVal Alignment As System.Boolean, _
    ByVal TwistCtrlOption As System.Short, _
    ByVal KeepTangency As System.Boolean, _
    ByVal ForceNonRational As System.Boolean, _
    ByVal StartMatchingType As System.Short, _
    ByVal EndMatchingType As System.Short, _
    ByVal IsThinBody As System.Boolean, _
    ByVal Thickness1 As System.Double, _

```

```

    ByVal Thickness2 As System.Double, _
    ByVal ThinType As System.Short, _
    ByVal PathAlign As System.Short, _
    ByVal UseFeatScope As System.Boolean, _
    ByVal UseAutoSelect As System.Boolean _
) As Feature

```

Visual Basic (Usage)



```

Dim instance As IFeatureManager
Dim Propagate As System.Boolean
Dim Alignment As System.Boolean
Dim TwistCtrlOption As System.Short
Dim KeepTangency As System.Boolean
Dim ForceNonRational As System.Boolean
Dim StartMatchingType As System.Short
Dim EndMatchingType As System.Short
Dim IsThinBody As System.Boolean
Dim Thickness1 As System.Double
Dim Thickness2 As System.Double
Dim ThinType As System.Short
Dim PathAlign As System.Short
Dim UseFeatScope As System.Boolean
Dim UseAutoSelect As System.Boolean
Dim value As Feature

```

```

value = instance.InsertCutSwept2(Propagate, Alignment, TwistCtrlOption, KeepTangency,
ForceNonRational, StartMatchingType, EndMatchingType, IsThinBody, Thickness1, Thickness2,
ThinType, PathAlign, UseFeatScope, UseAutoSelect)

```

C#

```

Feature InsertCutSwept2(
    System.bool Propagate,
    System.bool Alignment,
    System.short TwistCtrlOption,
    System.bool KeepTangency,
    System.bool ForceNonRational,
    System.short StartMatchingType,
    System.short EndMatchingType,
    System.bool IsThinBody,
    System.double Thickness1,
    System.double Thickness2,
    System.short ThinType,
    System.short PathAlign,
    System.bool UseFeatScope,
    System.bool UseAutoSelect
)

```

C++/CLI

```

Feature^ InsertCutSwept2(
    & System.bool Propagate,
    & System.bool Alignment,
    & System.short TwistCtrlOption,
    & System.bool KeepTangency,
    & System.bool ForceNonRational,
    & System.short StartMatchingType,
    & System.short EndMatchingType,
    & System.bool IsThinBody,
    & System.double Thickness1,
    & System.double Thickness2,
    & System.short ThinType,
    & System.short PathAlign,
    & System.bool UseFeatScope,
    & System.bool UseAutoSelect
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Propagate

Alignment

TwistCtrlOption

KeepTangency

ForceNonRational

StartMatchingType

EndMatchingType

IsThinBody

Thickness1

Thickness2

ThinType

PathAlign

UseFeatScope

UseAutoSelect

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::InsertCutSwept2](#).

See Also

[IFeatureManager Interface](#)

[IFeatureManager Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

InsertCutSwept3 Method (IFeatureManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertCutSwept3 Method (IFeatureManager)

Obsolete. Superseded by [IFeatureManager::InsertCutSwept4](#).

.NET Syntax

Visual Basic (Declaration)

```
Function InsertCutSwept3( _  
    ByVal Propagate As System.Boolean, _  
    ByVal Alignment As System.Boolean, _  
    ByVal TwistCtrlOption As System.Short, _  
    ByVal KeepTangency As System.Boolean, _  
    ByVal BAdvancedSmoothing As System.Boolean, _  
    ByVal StartMatchingType As System.Short, _  
    ByVal EndMatchingType As System.Short, _  
    ByVal IsThinBody As System.Boolean, _  
    ByVal Thickness1 As System.Double, _  
    ByVal Thickness2 As System.Double, _  
    ByVal ThinType As System.Short, _  
    ByVal PathAlign As System.Short, _  
    ByVal UseFeatScope As System.Boolean, _  
    ByVal UseAutoSelect As System.Boolean, _  
    ByVal TwistAngle As System.Double, _  
    ByVal BMergeSmoothFaces As System.Boolean _  
) As Feature
```

Visual Basic (Usage)

```
Dim instance As IFeatureManager  
Dim Propagate As System.Boolean  
Dim Alignment As System.Boolean  
Dim TwistCtrlOption As System.Short  
Dim KeepTangency As System.Boolean
```



```

Dim BAdvancedSmoothing As System.Boolean
Dim StartMatchingType As System.Short
Dim EndMatchingType As System.Short
Dim IsThinBody As System.Boolean
Dim Thickness1 As System.Double
Dim Thickness2 As System.Double
Dim ThinType As System.Short
Dim PathAlign As System.Short
Dim UseFeatScope As System.Boolean
Dim UseAutoSelect As System.Boolean
Dim TwistAngle As System.Double
Dim BMergeSmoothFaces As System.Boolean
Dim value As Feature

```

```

value = instance.InsertCutSwept3(Propagate, Alignment, TwistCtrlOption, KeepTangency,
BAdvancedSmoothing, StartMatchingType, EndMatchingType, IsThinBody, Thickness1, Thickness2,
ThinType, PathAlign, UseFeatScope, UseAutoSelect, TwistAngle, BMergeSmoothFaces)

```

C#

```

Feature InsertCutSwept3(
    System.bool Propagate,
    System.bool Alignment,
    System.short TwistCtrlOption,
    System.bool KeepTangency,
    System.bool BAdvancedSmoothing,
    System.short StartMatchingType,
    System.short EndMatchingType,
    System.bool IsThinBody,
    System.double Thickness1,
    System.double Thickness2,
    System.short ThinType,
    System.short PathAlign,
    System.bool UseFeatScope,
    System.bool UseAutoSelect,
    System.double TwistAngle,
    System.bool BMergeSmoothFaces
)

```

C++/CLI

```

Feature^ InsertCutSwept3(
    & System.bool Propagate,
    & System.bool Alignment,
    & System.short TwistCtrlOption,
    & System.bool KeepTangency,
    & System.bool BAdvancedSmoothing,
    & System.short StartMatchingType,
    & System.short EndMatchingType,
    & System.bool IsThinBody,
    & System.double Thickness1,
    & System.double Thickness2,
    & System.short ThinType,
    & System.short PathAlign,
    & System.bool UseFeatScope,
    & System.bool UseAutoSelect,
    & System.double TwistAngle,
    & System.bool BMergeSmoothFaces
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Propagate

True propagates the swept cut to the next edge, false causes the swept cut to occur only on the selected edge; to propagate to the next edge, the next edge must be tangent to the current edge

Alignment

If the curve used to sweep goes from one face to another or from one edge to another, then passing True causes the sweep to cut completely through the end faces of the cut, false causes the swept cut to begin and end perpendicular to the sweep curve; therefore, it may not break through the two end faces of the body being cut

TwistCtrlOption

Twist control options as defined by [swTwistControlType_e](#)

KeepTangency

If the sweep section has tangent segments, True to cause the corresponding surfaces in the resulting sweep to be tangent, false to not

BAdvancedSmoothing

If the sweep section has circular or elliptical arcs, true to approximate the sections and smooth the surfaces, false to not

StartMatchingType
 Tangency type as defined by [swTangencyType_e](#)
EndMatchingType
 Tangency type as defined by [swTangencyType_e](#)
IsThinBody
 True if this feature is a thin body, false if not
Thickness1
 Thickness value for the first direction
Thickness2
 Thickness value for the second direction
ThinType
 Thin wall type as defined by [swThinWallType_e](#)
PathAlign
 Align path type (see **Remarks**)
UseFeatScope
 True if the feature only affects selected bodies, false if the feature affects all bodies
UseAutoSelect
 True to automatically select all bodies and have the feature affect those bodies, false to select the bodies the feature affects (see **Remarks**)
TwistAngle
 If twistCtrlOption set to swTwistControlConstantTwistAlongPath, then specify end twist angle
BMergeSmoothFaces
 True to merge smooth faces, false to not

Return Value

Pointer to the [IFeature](#) object

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::InsertCutSwept3](#).

Remarks

Use [IModelDocExtension::SelectByID2](#) to select the profile and sweep curves. The mark for:

- 1 = Profile selection
- 2 = Guide-curve selection, if provided
- 4 = Sweep path

The PathAlign argument is available when TwistCtrlOption is set to 0 and can take one of these values:

- 0 = None; no correction (default)
- 2 = Direction vector; a plane, planar face, or line defines the path
- 3 = All faces; includes neighboring faces

When UseAutoSelect is false, the user must select the bodies that the feature will affect.

When using cut or cavity features that result in multiple bodies, you cannot select to keep all of the resulting bodies or one or more selected bodies.

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[ISweepFeatureData Interface](#)
[IFeatureManager::InsertProtrusionSwept3 Method](#)

Availability

SOLIDWORKS 2005 FCS, Revision Number 13.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertCutSwept4 Method (IFeatureManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertCutSwept4 Method (IFeatureManager)

Obsolete. Superseded by [IFeatureManager::InsertCutSwept5](#).

.NET Syntax

Visual Basic (Declaration)

```

Function InsertCutSwept4( _
    ByVal Propagate As System.Boolean, _
    ByVal Alignment As System.Boolean, _
    ByVal TwistCtrlOption As System.Short, _

```

```

ByVal KeepTangency As System.Boolean, _
ByVal BAdvancedSmoothing As System.Boolean, _
ByVal StartMatchingType As System.Short, _
ByVal EndMatchingType As System.Short, _
ByVal IsThinBody As System.Boolean, _
ByVal Thickness1 As System.Double, _
ByVal Thickness2 As System.Double, _
ByVal ThinType As System.Short, _
ByVal PathAlign As System.Short, _
ByVal UseFeatScope As System.Boolean, _
ByVal UseAutoSelect As System.Boolean, _
ByVal TwistAngle As System.Double, _
ByVal BMergeSmoothFaces As System.Boolean, _
ByVal AssemblyFeatureScope As System.Boolean, _
ByVal AutoSelectComponents As System.Boolean, _
ByVal PropagateFeatureToParts As System.Boolean _
) As Feature

```

Visual Basic (Usage)



```

Dim instance As IFeatureManager
Dim Propagate As System.Boolean
Dim Alignment As System.Boolean
Dim TwistCtrlOption As System.Short
Dim KeepTangency As System.Boolean
Dim BAdvancedSmoothing As System.Boolean
Dim StartMatchingType As System.Short
Dim EndMatchingType As System.Short
Dim IsThinBody As System.Boolean
Dim Thickness1 As System.Double
Dim Thickness2 As System.Double
Dim ThinType As System.Short
Dim PathAlign As System.Short
Dim UseFeatScope As System.Boolean
Dim UseAutoSelect As System.Boolean
Dim TwistAngle As System.Double
Dim BMergeSmoothFaces As System.Boolean
Dim AssemblyFeatureScope As System.Boolean
Dim AutoSelectComponents As System.Boolean
Dim PropagateFeatureToParts As System.Boolean
Dim value As Feature

```

```

value = instance.InsertCutSwept4(Propagate, Alignment, TwistCtrlOption, KeepTangency,
BAdvancedSmoothing, StartMatchingType, EndMatchingType, IsThinBody, Thickness1, Thickness2,
ThinType, PathAlign, UseFeatScope, UseAutoSelect, TwistAngle, BMergeSmoothFaces,
AssemblyFeatureScope, AutoSelectComponents, PropagateFeatureToParts)

```

C#

```

Feature InsertCutSwept4(
    System.bool Propagate,
    System.bool Alignment,
    System.short TwistCtrlOption,
    System.bool KeepTangency,
    System.bool BAdvancedSmoothing,
    System.short StartMatchingType,
    System.short EndMatchingType,
    System.bool IsThinBody,
    System.double Thickness1,
    System.double Thickness2,
    System.short ThinType,
    System.short PathAlign,
    System.bool UseFeatScope,
    System.bool UseAutoSelect,
    System.double TwistAngle,
    System.bool BMergeSmoothFaces,
    System.bool AssemblyFeatureScope,
    System.bool AutoSelectComponents,

```

```
        System.bool PropagateFeatureToParts
    )
```

C++/CLI

```
Feature^ InsertCutSwept4(
& System.bool Propagate,
& System.bool Alignment,
& System.short TwistCtrlOption,
& System.bool KeepTangency,
& System.bool BAdvancedSmoothing,
& System.short StartMatchingType,
& System.short EndMatchingType,
& System.bool IsThinBody,
& System.double Thickness1,
& System.double Thickness2,
& System.short ThinType,
& System.short PathAlign,
& System.bool UseFeatScope,
& System.bool UseAutoSelect,
& System.double TwistAngle,
& System.bool BMergeSmoothFaces,
& System.bool AssemblyFeatureScope,
& System.bool AutoSelectComponents,
& System.bool PropagateFeatureToParts
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Propagate

True propagates the sweep cut to the next edge, false causes the sweep cut to occur only on the selected edge; to propagate to the next edge, the next edge must be tangent to the current edge

Alignment

If the curve used to sweep goes from one face to another or from one edge to another, then true causes the sweep to cut completely through the end faces of the cut, and false causes the cut to begin and end perpendicular to the sweep curve; therefore, it may not break through the two end faces of the body being cut

TwistCtrlOption

Twist control options as defined in [swTwistControlType_e](#)

KeepTangency

If the sweep section has tangent segments, true causes the corresponding surfaces in the resulting sweep to be tangent, false does not

BAdvancedSmoothing

If the sweep section has circular or elliptical arcs, true approximates the sections and smooths the surfaces, false does not

StartMatchingType

Tangency type as defined in [swTangencyType_e](#)

EndMatchingType

Tangency type as defined in [swTangencyType_e](#)

IsThinBody

True if this feature is a thin body, false if not

Thickness1

Thickness value for the first direction

Thickness2

Thickness value for the second direction

ThinType

Thin wall type as defined in [swThinWallType_e](#)

PathAlign

Align path type (see **Remarks**)

UseFeatScope

True if the feature only affects selected bodies, false if the feature affects all bodies

UseAutoSelect

True to automatically select all bodies and have the feature affect those bodies, false to select the bodies the feature affects (see **Remarks**)

TwistAngle

If TwistCtrlOption is set to swTwistControlType_e.swTwistControlConstantTwistAlongPath, then specify end twist angle

BMergeSmoothFaces

True to merge smooth faces, false to not

AssemblyFeatureScope

True if the sweep cut affects only selected components in the assembly, false if the sweep cut affects all components in the assembly (see **Remarks**)

AutoSelectComponents

True to auto-select all affected components in the assembly, false to use manually selected components (see **Remarks**)

PropagateFeatureToParts

True to extend the sweep cut feature to all affected parts in the assembly, false to just insert the sweep cut into the assembly (see **Remarks**)

Return Value

Pointer to the [IFeature](#) object

Visual Basic for Applications (VBA) Syntax

See [FeatureManager.InsertCutSwept4](#).

Example

[Insert Sweep Cut Feature \(C#\)](#)
[Insert Sweep Cut Feature \(VB.NET\)](#)
[Insert Sweep Cut Feature \(VBA\)](#)

Remarks

Before calling this method, call [IModelDocExtension::SelectByID2](#) multiple times to select the profile, guide curves, and sweep path for the cut, setting the Mark parameter in each case as follows:

- 1 = Profile selection
- 2 = Guide-curve selection, if provided
- 4 = Sweep path selection

The PathAlign argument is available when TwistCtrlOption is set to 0 and can take one of these values:

- 0 = None; no correction (default)
- 2 = Direction vector; a plane, planar face, or line defines the path
- 3 = All faces; includes neighboring faces

When UseAutoSelect is false, the user must select the bodies that the feature will affect.

When using cut or cavity features that result in multiple bodies, you cannot select to keep all of the resulting bodies or one or more selected bodies.

Use AssemblyFeatureScope, AutoSelectComponents, and PropagateFeatureToParts to insert sweep cuts into an assembly. AssemblyFeatureScope and AutoSelectComponents perform just like the configuration of the Feature Scope section on the PropertyManager page of the sweep feature:

AssemblyFeatureScope setting	AutoSelectComponents setting	PropertyManager page Feature Scope setting
False	Ignored	All components selected Auto-select not visible
True	If true, affected components are automatically selected If false, manually select the affected components in the view before calling this API	Selected components selected If Auto-select is not selected, then manually select affected components in the view

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[ISweepFeatureData Interface](#)
[IFeatureManager::InsertProtrusionSwept3 Method](#)
[IFeatureManager::InsertSweepSurface2 Method \(\)](#)

Availability

SOLIDWORKS 2011 FCS, Revision Number 19.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertCutSwept5 Method (IFeatureManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertCutSwept5 Method (IFeatureManager)

Obsolete. See [Remarks](#).

.NET Syntax

Visual Basic (Declaration)

```
Function InsertCutSwept5( _  
    ByVal Propagate As System.Boolean, _  
    ByVal Alignment As System.Boolean, _  
    ByVal TwistCtrlOption As System.Integer, _  
    ByVal KeepTangency As System.Boolean, _  
    ByVal AdvancedSmoothing As System.Boolean, _  
    ByVal StartMatchingType As System.Integer, _  
    ByVal EndMatchingType As System.Integer, _  
    ByVal IsThinBody As System.Boolean, _  
    ByVal Thickness1 As System.Double, _  
    ByVal Thickness2 As System.Double, _  
    ByVal ThinType As System.Integer, _
```

```

ByVal PathAlign As System.Integer, _
ByVal UseFeatScope As System.Boolean, _
ByVal UseAutoSelect As System.Boolean, _
ByVal TwistAngle As System.Double, _
ByVal BMergeSmoothFaces As System.Boolean, _
ByVal AssemblyFeatureScope As System.Boolean, _
ByVal AutoSelectComponents As System.Boolean, _
ByVal PropagateFeatureToParts As System.Boolean, _
ByVal CircularProfile As System.Boolean, _
ByVal CircularProfileDiameter As System.Double, _
ByVal Direction As System.Integer _
) As Feature

```

Visual Basic (Usage)



```

Dim instance As IFeatureManager
Dim Propagate As System.Boolean
Dim Alignment As System.Boolean
Dim TwistCtrlOption As System.Integer
Dim KeepTangency As System.Boolean
Dim BAdvancedSmoothing As System.Boolean
Dim StartMatchingType As System.Integer
Dim EndMatchingType As System.Integer
Dim IsThinBody As System.Boolean
Dim Thickness1 As System.Double
Dim Thickness2 As System.Double
Dim ThinType As System.Integer
Dim PathAlign As System.Integer
Dim UseFeatScope As System.Boolean
Dim UseAutoSelect As System.Boolean
Dim TwistAngle As System.Double
Dim BMergeSmoothFaces As System.Boolean
Dim AssemblyFeatureScope As System.Boolean
Dim AutoSelectComponents As System.Boolean
Dim PropagateFeatureToParts As System.Boolean
Dim CircularProfile As System.Boolean
Dim CircularProfileDiameter As System.Double
Dim Direction As System.Integer
Dim value As Feature

```

```

value = instance.InsertCutSwept5(Propagate, Alignment, TwistCtrlOption, KeepTangency,
BAdvancedSmoothing, StartMatchingType, EndMatchingType, IsThinBody, Thickness1, Thickness2,
ThinType, PathAlign, UseFeatScope, UseAutoSelect, TwistAngle, BMergeSmoothFaces,
AssemblyFeatureScope, AutoSelectComponents, PropagateFeatureToParts, CircularProfile,
CircularProfileDiameter, Direction)

```

C#

```

Feature InsertCutSwept5(
    System.bool Propagate,
    System.bool Alignment,
    System.int TwistCtrlOption,
    System.bool KeepTangency,
    System.bool BAdvancedSmoothing,
    System.int StartMatchingType,
    System.int EndMatchingType,
    System.bool IsThinBody,
    System.double Thickness1,
    System.double Thickness2,
    System.int ThinType,
    System.int PathAlign,
    System.bool UseFeatScope,
    System.bool UseAutoSelect,
    System.double TwistAngle,
    System.bool BMergeSmoothFaces,
    System.bool AssemblyFeatureScope,
    System.bool AutoSelectComponents,
    System.bool PropagateFeatureToParts,
    System.bool CircularProfile,

```

```

        System.double CircularProfileDiameter,
        System.int Direction
    )

```

C++/CLI

```

Feature^ InsertCutSwept5(
& System.bool Propagate,
& System.bool Alignment,
& System.int TwistCtrlOption,
& System.bool KeepTangency,
& System.bool BAdvancedSmoothing,
& System.int StartMatchingType,
& System.int EndMatchingType,
& System.bool IsThinBody,
& System.double Thickness1,
& System.double Thickness2,
& System.int ThinType,
& System.int PathAlign,
& System.bool UseFeatScope,
& System.bool UseAutoSelect,
& System.double TwistAngle,
& System.bool BMergeSmoothFaces,
& System.bool AssemblyFeatureScope,
& System.bool AutoSelectComponents,
& System.bool PropagateFeatureToParts,
& System.bool CircularProfile,
& System.double CircularProfileDiameter,
& System.int Direction
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Propagate

True propagates the sweep cut to the next edge, false causes the sweep cut to occur only on the selected edge; to propagate to the next edge, the next edge must be tangent to the current edge

Alignment

If the curve used to sweep goes from one face to another or from one edge to another, then true causes the sweep to cut completely through the end faces of the cut, and false causes the cut to begin and end perpendicular to the sweep curve; thus, it cannot break through the two end faces of the body being cut

TwistCtrlOption

Twist control options as defined in [swTwistControlType_e](#)

KeepTangency

If the sweep section has tangent segments, true to cause the corresponding surfaces in the resulting sweep to be tangent, false to not

BAdvancedSmoothing

If the sweep section has circular or elliptical arcs, true to approximate the sections and smooth the surfaces, false to not

StartMatchingType

Tangency type as defined in [swTangencyType_e](#)

EndMatchingType

Tangency type as defined in [swTangencyType_e](#)

IsThinBody

True if this feature is a thin body, false if not

Thickness1

Thickness value for the first direction

Thickness2

Thickness value for the second direction

ThinType

Thin wall type as defined in [swThinWallType_e](#)

PathAlign

Align path type (see **Remarks**)

UseFeatScope

True if the feature only affects selected bodies, false if the feature affects all bodies

UseAutoSelect

True to automatically select all bodies and have the feature affect those bodies, false to select the bodies the feature affects (see **Remarks**)

TwistAngle

If [TwistCtrlOption](#) set to [swTwistControlType_e.swTwistControlConstantTwistAlongPath](#), then specify end twist angle

BMergeSmoothFaces

True to merge smooth faces, false to not

AssemblyFeatureScope

True if the sweep cut affects only selected components in the assembly, false if the sweep cut affects all components in the assembly (see **Remarks**)

AutoSelectComponents

True to auto-select all affected components in the assembly, false to use manually selected components (see **Remarks**)

PropagateFeatureToParts

True to extend the sweep cut feature to all affected parts in the assembly, false to just insert the sweep cut into the assembly (see **Remarks**)

CircularProfile

True to use a circular profile, false to use the selected sketch profile or solid body

CircularProfileDiameter

If [CircularProfile](#) is true, then specify the diameter of the circular profile

Direction

Direction as defined in [swSweepDirection_e](#) (see **Remarks**)

Return Value

[Feature](#)

See [FeatureManager.InsertCutSweep5](#).

 Example

[Create Cut-sweep Feature Using Tool Body \(C#\)](#)
[Create Cut-sweep Feature Using Tool Body \(VB.NET\)](#)
[Create Cut-sweep Feature Using Tool Body \(VBA\)](#)
[Create Cut-sweep Feature Using Circular Profile \(C#\)](#)
[Create Cut-sweep Feature Using Circular Profile \(VB.NET\)](#)
[Create Cut-sweep Feature Using Circular Profile \(VBA\)](#)

 Remarks

SOLIDWORKS 2018 introduces a new sweep architecture, making this method obsolete. See [Sweep Features and SweepFeatureData Objects](#) to create this cut-sweep feature.

Before calling this method, call [IModelDocExtension::SelectByID2](#) multiple times to select the sketch profile or tool body, guide curves, and sweep path for the cut. Specify these marks:

- 1 = Sketch profile or tool body
- 2 = Guide curve, if provided
- 4 = Sweep path

The PathAlign argument is available when TwistCtrlOption is set to swTwistControlType_e.swTwistControlFollowPath and can take one of these values:

- 0 = None; no correction (default)
- 2 = Direction vector; a plane, planar face, or line defines the path
- 3 = All faces; includes neighboring faces

When UseAutoSelect is false, the user must select the bodies that the feature will affect.

When using cut or cavity features that result in multiple bodies, you cannot select to keep all of the resulting bodies or one or more selected bodies.

Use AssemblyFeatureScope, AutoSelectComponents, and PropagateFeatureToParts to insert sweep cuts into an assembly. AssemblyFeatureScope and AutoSelectComponents perform just like the configuration of the Feature Scope section on the PropertyManager page of the sweep feature:

AssemblyFeatureScope setting	AutoSelectComponents setting	PropertyManager page Feature Scope setting
False	Ignored	• All components selected • Auto-select not visible
True	If true, affected components are automatically selected If false, manually select the affected components in the view before calling this method	Selected components selected If Auto-select is not selected, then manually select affected components in the view

Direction only applies to sketch profiles and only when the sketch profile is not coincident with an end of the path.

 See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[ISweepFeatureData Interface](#)
[IFeatureManager.InsertProtrusionSweep4 Method \(\)](#)
[IFeatureManager.InsertSweepSurface3 Method \(\)](#)
[ISweepFeatureData Interface](#)

 Availability

SOLIDWORKS 2017 FCS, Revision Number 25.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertDeleteBody Method (IFeatureManager)

See Also



SolidWorks.Interop.sldworks Namespace > IFeatureManager Interface : InsertDeleteBody Method (IFeatureManager)

Obsolete. Superseded by [IFeatureManager.InsertDeleteBody2](#).

 .NET Syntax

Visual Basic (Declaration)

Function InsertDeleteBody() As Feature

Visual Basic (Usage)



```
Dim instance As IFeatureManager
Dim value As Feature

value = instance.InsertDeleteBody()
```

C#

```
Feature InsertDeleteBody()
```

C++/CLI

```
Feature^ InsertDeleteBody();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Pointer to the [IFeature](#) object

Visual Basic for Applications (VBA) Syntax

See [FeatureManager.InsertDeleteBody](#).

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[IDeleteBodyFeatureData Interface](#)

Availability

SOLIDWORKS 2003 FCS, Revision Number 11.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

InsertDeleteBody2 Method (IFeatureManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertDeleteBody2 Method (IFeatureManager)

Inserts a Body-Delete/Keep feature.

.NET Syntax

Visual Basic (Declaration)

```
Function InsertDeleteBody2( _
    ByVal KeepBodies As System.Boolean _
) As Feature
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager
Dim KeepBodies As System.Boolean
Dim value As Feature

value = instance.InsertDeleteBody2(KeepBodies)
```

C#

```
Feature InsertDeleteBody2(
    System.bool KeepBodies
)
```

C++/CLI

```
Feature^ InsertDeleteBody2(  
& System.bool KeepBodies  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

KeepBodies
True to keep bodies, false to delete bodies

Return Value

[IFeature](#)

Visual Basic for Applications (VBA) Syntax

See [FeatureManager.InsertDeleteBody2](#).

Example

[Insert Body-Delete/Keep Feature \(VBA\)](#)
[Insert Body-Delete/Keep Feature \(VB.NET\)](#)
[Insert Body-Delete/Keep Feature \(C#\)](#)

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[IDeleteBodyFeatureData Interface](#)

Availability

SOLIDWORKS 2015 FCS, Revision Number 23.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertDeleteHoleForSurface Method (IFeatureManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertDeleteHoleForSurface Method (IFeatureManager)

Inserts a Delete Hole feature for one or more selected hole edges on a surface.

.NET Syntax

Visual Basic (Declaration)

```
Function InsertDeleteHoleForSurface() As Feature
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager  
Dim value As Feature  
  
value = instance.InsertDeleteHoleForSurface()
```

C#

```
Feature InsertDeleteHoleForSurface()
```

C++/CLI

```
Feature^ InsertDeleteHoleForSurface();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

[IFeature](#)

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::InsertDeleteHoleForSurface](#).

Remarks

Before calling this method, call [IModelDocExtension::SelectByID2](#) to select the hole edges to remove from the surface.

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)

Availability

SOLIDWORKS 2019 FCS, Revision Number 27.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertDerivedPattern Method (IFeatureManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertDerivedPattern Method (IFeatureManager)

Obsolete. Superseded by [IFeatureManager::InsertDerivedPattern2](#).

.NET Syntax

Visual Basic (Declaration)

Function InsertDerivedPattern() As Feature

Visual Basic (Usage)



```
Dim instance As IFeatureManager
Dim value As Feature
```

```
value = instance.InsertDerivedPattern()
```

C#

```
Feature InsertDerivedPattern()
```

C++/CLI

```
Feature^ InsertDerivedPattern();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Derived pattern [feature](#)

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::InsertDerivedPattern](#).

Remarks

Use [IModelDocExtension::SelectByID2](#) to select the components, which must be ordered:

- seed component = 1
- pattern feature = 2

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[IDerivedPatternFeatureData Interface](#)

Availability

InsertDerivedPattern2 Method (IFeatureManager)

[See Also](#)[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertDerivedPattern2 Method (IFeatureManager)**Obsolete.** See [IFeatureManager::CreateFeature](#) and the Remarks in [IDerivedPatternFeatureData](#).

.NET Syntax

Visual Basic (Declaration)

Function InsertDerivedPattern2() As Feature

Visual Basic (Usage)

Dim instance As [IFeatureManager](#)

Dim value As Feature

value = instance.InsertDerivedPattern2()

C#

Feature InsertDerivedPattern2()

C++/CLI

Feature^ InsertDerivedPattern2();

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).**Return Value**Derived pattern [feature](#)

Visual Basic for Applications (VBA) Syntax

See [FeatureManager.InsertDerivedPattern2](#).

Remarks

Use [ModelDocExtension::SelectByID2](#) to select the seed components and pattern feature in any order.

- seed component = 1
- pattern feature = 2

NOTE: A derived pattern feature is also called a pattern driven pattern feature.

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)

Availability

SOLIDWORKS 2017 FCS, Revision Number 25.0

InsertDwgOrDxfFile Method (IFeatureManager)

[See Also](#)[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertDwgOrDxfFile Method (IFeatureManager)**Obsolete. Superseded by** [IFeatureManager::InsertDwgOrDxfFile2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function InsertDwgOrDxfFile( _  
    ByVal FileName As System.String _  
    ) As Feature
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager  
Dim FileName As System.String  
Dim value As Feature  
  
value = instance.InsertDwgOrDxfFile(FileName)
```

C#

```
Feature InsertDwgOrDxfFile(  
    System.string FileName  
)
```

C++/CLI

```
Feature^ InsertDwgOrDxfFile(  
& System.String^ FileName  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

FileName

Name of the DXF or DWG image file

Return Value

Pointer to the [IFeature](#) object

Visual Basic for Applications (VBA) Syntax

See [FeatureManager.InsertDwgOrDxfFile](#).

Remarks

This method returns:

- Nothing or null if the file contains solid bodies data.
- is not supported for use in assembly documents.

You must also select a plane or planar surface before calling this method.

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)

Availability

SOLIDWORKS 2003 FCS, Revision Number 11.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertDwgOrDxfFile2 Method (IFeatureManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertDwgOrDxfFile2 Method (IFeatureManager)

Inserts a DXF/DWG image feature.

.NET Syntax

Visual Basic (Declaration)

```
Function InsertDwgOrDxfFile2( _
    ByVal FileName As System.String, _
    ByVal DxfDwgImportData As System.Object _
) As Feature
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager
Dim FileName As System.String
Dim DxfDwgImportData As System.Object
Dim value As Feature

value = instance.InsertDwgOrDxfFile2(FileName, DxfDwgImportData)
```

C#

```
Feature InsertDwgOrDxfFile2(
    System.string FileName,
    System.object DxfDwgImportData
)
```

C++/CLI

```
Feature^ InsertDwgOrDxfFile2(
    & System.String^ FileName,
    & System.Object^ DxfDwgImportData
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

FileName

Name of the DXF/DWG image file

DxfDwgImportData
[IImportDxfDwgData](#)

Return Value

[IFeature](#)

Visual Basic for Applications (VBA) Syntax

See [FeatureManager.InsertDwgOrDxfFile2](#).

Example

[Insert and Position DXF/DWG File in Drawing \(VBA\)](#)
[Insert and Position DXF/DWG File in Drawing \(VB.NET\)](#)
[Insert and Position DXF/DWG File in Drawing \(C#\)](#)
[Insert DXF/DWG File and Add Dimensions \(VBA\)](#)
[Insert DXF/DWG File and Add Dimensions \(VB.NET\)](#)
[Insert DXF/DWG File and Add Dimensions \(C#\)](#)

Remarks

Before calling this method, select a plane or planar surface on which to place the image.

This method:

- returns Nothing or null if the file contains solid bodies data.
- is not supported for use in assembly documents.

When importing DXF/DWG data, you can:

- Let SOLIDWORKS determine the default values:
 - Paper size and sheet scale are computed to fit the input data.
 - Length unit is determined from the header of the input DXF/DWG file.
 - Sheet name is the same as the layout name in the input DXF/DWG file.

- or -

- Set your own values:
 1. Call [ISldWorks::GetImportFileData](#) to obtain the IImportDxfDwgData interface.

2. Use the following methods with a Sheet argument of "" (an empty string), to set up your defaults:

- [IImportDxfDwgData::GetPaperSize](#)
- [IImportDxfDwgData::GetPosition](#)
- [IImportDxfDwgData::GetSheetScale](#)
- [IImportDxfDwgData::ImportMethod](#)
- [IImportDxfDwgData::LengthUnit](#)
- [IImportDxfDwgData::SetPaperSize](#)
- [IImportDxfDwgData::SetPosition](#)
- [IImportDxfDwgData::SetSheetScale](#)
- [IImportDxfDwgData::SheetName](#)

NOTES:

- Getting the IImportDxfDwgData interface does not get default values from the input file. Any values not set by you are set to the values computed by SOLIDWORKS.
- If the DWG/DXF file has multiple sheets, use these methods with a valid layout name in the Sheet argument to set up sheet specific settings, which override the default settings. If any of the individual items are not specified for a given layout name, then the value used is from the defaults (layout name ""). If the default value is not specified, then SOLIDWORKS computes and uses a meaningful value for that item.

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[ISldWorks::LoadFile4 Method](#)

Availability

SOLIDWORKS 2012 FCS, Revision Number 20.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertEdgeMerge Method (IFeatureManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertEdgeMerge Method (IFeatureManager)

Merges multiple edges into a single edge using the selected faces when importing data.

.NET Syntax

Visual Basic (Declaration)

```
Function InsertEdgeMerge( _  
    ByVal Angular\_tolerance As System.Double, _  
    ByVal Edge\_length\_tolerance As System.Double _  
) As Feature
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager  
Dim Angular_tolerance As System.Double  
Dim Edge_length_tolerance As System.Double  
Dim value As Feature  
  
value = instance.InsertEdgeMerge(Angular_tolerance, Edge_length_tolerance)
```

C#

```
Feature InsertEdgeMerge(  
    System.double Angular\_tolerance,  
    System.double Edge\_length\_tolerance  
)
```

C++/CLI

```
Feature^ InsertEdgeMerge(  
& System.double Angular\_tolerance,
```

```
& System.double Edge\_length\_tolerance  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Angular_tolerance
Angular tolerance
Edge_length_tolerance
Edge length tolerance

Return Value

Pointer to the [IFeature](#) object

Visual Basic for Applications (VBA) Syntax

See [FeatureManager.InsertEdgeMerge](#).

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)

Availability

SOLIDWORKS 2005 FCS, Revision Number 13.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertEndCapFeature Method (IFeatureManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertEndCapFeature Method (IFeatureManager)

Obsolete. Superseded by [IFeatureManager.InsertEndCapFeature3](#).

.NET Syntax

Visual Basic (Declaration)

```
Function InsertEndCapFeature( _  
    ByVal Depth As System.Double, _  
    ByVal BIsGivenOffset As System.Boolean, _  
    ByVal BIsChamfer As System.Boolean, _  
    ByVal OffsetValue As System.Double, _  
    ByVal WallThicknessRatio As System.Double, _  
    ByVal ChamferValue As System.Double _  
) As Feature
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager  
Dim Depth As System.Double  
Dim BIsGivenOffset As System.Boolean  
Dim BIsChamfer As System.Boolean  
Dim OffsetValue As System.Double  
Dim WallThicknessRatio As System.Double  
Dim ChamferValue As System.Double  
Dim value As Feature
```

```
value = instance.InsertEndCapFeature(Depth, BIsGivenOffset, BIsChamfer, OffsetValue,  
WallThicknessRatio, ChamferValue)
```

C#

```
Feature InsertEndCapFeature(  
    System.double Depth,  
    System.bool BIsGivenOffset,  
    System.bool BIsChamfer,  
    System.double OffsetValue,
```

```

        System.double WallThicknessRatio,
        System.double ChamferValue
    )

```

C++/CLI

```

Feature^ InsertEndCapFeature(
&    System.double Depth,
&    System.bool BIsGivenOffset,
&    System.bool BIsChamfer,
&    System.double OffsetValue,
&    System.double WallThicknessRatio,
&    System.double ChamferValue
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Depth

Thickness of the end cap

BIsGivenOffset

True to provide an offset value, false to provide a thickness ratio

BIsChamfer

True if end cap feature is chamfered, false if end cap is filleted

OffsetValue

Edge offset value; valid only if BIsGivenOffset is true

WallThicknessRatio

Wall thickness ratio; valid only if BIsGivenOffset is false

ChamferValue

Chamfer distance if BIsChamfer is true, fillet radius if BIsChamfer is false

Return Value

[IFeature](#)

Visual Basic for Applications (VBA) Syntax

See [FeatureManager.InsertEndCapFeature](#).

See Also

[IFeatureManager Interface](#)

[IFeatureManager Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertEndCapFeature2 Method (IFeatureManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertEndCapFeature2 Method (IFeatureManager)

Inserts an end cap feature using the specified end faces of a structural member.

.NET Syntax

Visual Basic (Declaration)

```

Function InsertEndCapFeature2( _
    ByVal Depth As System.Double, _
    ByVal BIsGivenOffset As System.Boolean, _
    ByVal BIsChamfer As System.Boolean, _
    ByVal OffsetValue As System.Double, _
    ByVal WallThicknessRatio As System.Double, _
    ByVal ChamferValue As System.Double, _
    ByVal Faces As System.Object _
) As Feature

```

Visual Basic (Usage)

```

Dim instance As IFeatureManager
Dim Depth As System.Double
Dim BIsGivenOffset As System.Boolean
Dim BIsChamfer As System.Boolean
Dim OffsetValue As System.Double

```



```
Dim WallThicknessRatio As System.Double
Dim ChamferValue As System.Double
Dim Faces As System.Object
Dim value As Feature
```

```
value = instance.InsertEndCapFeature2(Depth, BIsGivenOffset, BIsChamfer, OffsetValue,
WallThicknessRatio, ChamferValue, Faces)
```

C#

```
Feature InsertEndCapFeature2(
    System.double Depth,
    System.bool BIsGivenOffset,
    System.bool BIsChamfer,
    System.double OffsetValue,
    System.double WallThicknessRatio,
    System.double ChamferValue,
    System.object Faces
)
```

C++/CLI

```
Feature^ InsertEndCapFeature2(
    & System.double Depth,
    & System.bool BIsGivenOffset,
    & System.bool BIsChamfer,
    & System.double OffsetValue,
    & System.double WallThicknessRatio,
    & System.double ChamferValue,
    & System.Object^ Faces
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Depth
Depth of the end cap
BIsGivenOffset
True if end cap is offset, false if not
BIsChamfer
True if end cap feature is chamfered, false if not
OffsetValue
Value by which to offset the end cap
WallThicknessRatio
Wall thickness ratio
ChamferValue
Angle of the chamfer
Faces
Array of [Face2](#) objects

Return Value

[IFeature](#)

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::InsertEndCapFeature2](#).

Example

[Insert Weldment Features \(VBA\)](#)
[Insert Weldment Features \(VB.NET\)](#)
[Insert Weldment Features \(C#\)](#)

Remarks

Call [IFeatureManager::InsertEndCapFeature3](#) if you want to pre-select the faces to end-cap in the graphics area.

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[IEndCapFeatureData Interface](#)

Availability

SOLIDWORKS 2010 FCS, Revision Number 18.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertEndCapFeature3 Method (IFeatureManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertEndCapFeature3 Method (IFeatureManager)

Inserts an end cap feature for one or more pre-selected open ends of a structural member.

Visual Basic Syntax

Visual Basic (Declaration)

```
Function InsertEndCapFeature3( _  
    ByVal Depth As System.Double, _  
    ByVal BIsGivenOffset As System.Boolean, _  
    ByVal BIsChamfer As System.Boolean, _  
    ByVal OffsetValue As System.Double, _  
    ByVal WallThicknessRatio As System.Double, _  
    ByVal ChamferValue As System.Double, _  
    ByVal BIsCornerTreatment As System.Boolean, _  
    ByVal DepthOffset As System.Double, _  
    ByVal BIsReverse As System.Boolean, _  
    ByVal BIsEndCapInward As System.Integer _  
) As Feature
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager  
Dim Depth As System.Double  
Dim BIsGivenOffset As System.Boolean  
Dim BIsChamfer As System.Boolean  
Dim OffsetValue As System.Double  
Dim WallThicknessRatio As System.Double  
Dim ChamferValue As System.Double  
Dim BIsCornerTreatment As System.Boolean  
Dim DepthOffset As System.Double  
Dim BIsReverse As System.Boolean  
Dim BIsEndCapInward As System.Integer  
Dim value As Feature
```

```
value = instance.InsertEndCapFeature3(Depth, BIsGivenOffset, BIsChamfer, OffsetValue,  
WallThicknessRatio, ChamferValue, BIsCornerTreatment, DepthOffset, BIsReverse, BIsEndCapInward)
```

C#

```
Feature InsertEndCapFeature3(  
    System.double Depth,  
    System.bool BIsGivenOffset,  
    System.bool BIsChamfer,  
    System.double OffsetValue,  
    System.double WallThicknessRatio,  
    System.double ChamferValue,  
    System.bool BIsCornerTreatment,  
    System.double DepthOffset,  
    System.bool BIsReverse,  
    System.int BIsEndCapInward  
)
```

C++/CLI

```
Feature^ InsertEndCapFeature3(  
& System.double Depth,  
& System.bool BIsGivenOffset,  
& System.bool BIsChamfer,  
& System.double OffsetValue,
```

```
& System.double WallThicknessRatio,  
& System.double ChamferValue,  
& System.bool BIsCornerTreatment,  
& System.double DepthOffset,  
& System.bool BIsReverse,  
& System.int BIsEndCapInward  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Depth

Thickness of the end cap

BIsGivenOffset

True to provide an offset value, false to provide a thickness ratio

BIsChamfer

True if end cap feature is chamfered, false if end cap is filleted

OffsetValue

Edge offset value; valid only if BIsGivenOffset is true

WallThicknessRatio

Wall thickness ratio; valid only if BIsGivenOffset is false

ChamferValue

Chamfer distance if BIsChamfer is true, fillet radius if BIsChamfer is false

BIsCornerTreatment

True to chamfer or fillet the end cap corners, false to not; valid only if BIsGivenOffset is false

DepthOffset

Inset distance; valid only if BIsEndCapInward = 2

BIsReverse

True to reverse the offset or thickness ratio, false to not

BIsEndCapInward

Thickness direction as defined in [swEndCapThicknessDirection_e](#)

Return Value

[IFeature](#)

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::InsertEndCapFeature3](#).

Example

[Insert Weldment End Cap \(VBA\)](#)

[Insert Weldment End Cap \(VB.NET\)](#)

[Insert Weldment End Cap \(C#\)](#)

Remarks

Before calling this method, select one or more end [faces](#) of a structural member in the graphics area.

Instead of using this method, you can pass the faces in an argument array of [IFeatureManager::InsertEndCapFeature2](#).

See Also

[IFeatureManager Interface](#)

[IFeatureManager Members](#)

[IEndCapFeatureData Interface](#)

Availability

SOLIDWORKS 2015 FCS, Revision Number 23.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

InsertFeatureChamfer Method (IFeatureManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertFeatureChamfer Method (IFeatureManager)

Inserts a chamfer.

.NET Syntax

Visual Basic (Declaration)

```

Function InsertFeatureChamfer( _
    ByVal Options As System.Integer, _
    ByVal ChamferType As System.Integer, _
    ByVal Width As System.Double, _
    ByVal Angle As System.Double, _
    ByVal OtherDist As System.Double, _
    ByVal VertexChamDist1 As System.Double, _
    ByVal VertexChamDist2 As System.Double, _
    ByVal VertexChamDist3 As System.Double _
) As Feature

```

Visual Basic (Usage)



```

Dim instance As IFeatureManager
Dim Options As System.Integer
Dim ChamferType As System.Integer
Dim Width As System.Double
Dim Angle As System.Double
Dim OtherDist As System.Double
Dim VertexChamDist1 As System.Double
Dim VertexChamDist2 As System.Double
Dim VertexChamDist3 As System.Double
Dim value As Feature

```

```

value = instance.InsertFeatureChamfer(Options, ChamferType, Width, Angle, OtherDist,
VertexChamDist1, VertexChamDist2, VertexChamDist3)

```

C#

```

Feature InsertFeatureChamfer(
    System.int Options,
    System.int ChamferType,
    System.double Width,
    System.double Angle,
    System.double OtherDist,
    System.double VertexChamDist1,
    System.double VertexChamDist2,
    System.double VertexChamDist3
)

```

C++/CLI

```

Feature^ InsertFeatureChamfer(
    & System.int Options,
    & System.int ChamferType,
    & System.double Width,
    & System.double Angle,
    & System.double OtherDist,
    & System.double VertexChamDist1,
    & System.double VertexChamDist2,
    & System.double VertexChamDist3
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Options

Options as defined by [swFeatureChamferOption_e](#)

ChamferType

Chamfer type as defined by [swChamferType_e](#)

Width

If ChamferType is swChamferAngleDistance, then specify width of chamfer

Angle

If ChamferType is swChamferAngleDistance, then specify the angle of the chamfer

OtherDist

If ChamferType is swChamferEqualDistance, then you can specify a single value so that all distances are equal

VertexChamDist1

If ChamferType is swChamferDistanceDistance or swChamferVertex, then specify a value for the distance on first side

VertexChamDist2

If ChamferType is swChamferDistanceDistance or swChamferVertex, then specify a value for the distance on second side

VertexChamDist3

If ChamferType is swChamferVertex, then specify a value for the distance on third side

Return Value

Pointer to the [IFeature](#) object

Visual Basic for Applications (VBA) Syntax

See [FeatureManager.InsertFeatureChamfer](#).

Example

See the [IChamferFeatureData2](#) examples.

Remarks

Both `swChamferType_e.swChamferAngleDistance` and `swChamferType_e.swChamferDistanceDistance` are edge chamfers. This means that all measurements are from edges. An angle-distance chamfer requires an angle and a distance; a distance-distance chamfer requires two distances, one for each side of the chamfered edge.

A `swChamferType_e.swChamferVertex` chamfer only works on a vertex with three adjacent edges of the same convexity. A vertex chamfer requires three distances along three adjacent edges. For non-linear edges, this value is an arc length value; it is not a chordal value. See [Vertex::EnumEdgesOriented](#) to determine the edge order used by this method.

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)

Availability

SOLIDWORKS 2005 FCS, Revision Number 13.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertFeatureLock Method (IFeatureManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertFeatureLock Method (IFeatureManager)

Locks or freezes a selected feature.

.NET Syntax

Visual Basic (Declaration)

```
Function InsertFeatureLock() As Feature
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager  
Dim value As Feature  
  
value = instance.InsertFeatureLock()
```

C#

```
Feature InsertFeatureLock()
```

C++/CLI

```
Feature^ InsertFeatureLock();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

[IFeature](#)

Visual Basic for Applications (VBA) Syntax

See [FeatureManager.InsertFeatureLock](#).

Example

```
'VBA
' Preconditions:
' 1. Open a part.
' 2. Select a feature.
' Postconditions: Observe that the selected feature is locked, and a gold freeze bar appears after the feature in the FeatureManager design tree.
```

Option Explicit

```
Dim swApp As SldWorks.SldWorks
Dim swModel As SldWorks.ModelDoc2
Dim swSelMgr As SldWorks.SelectionMgr
Dim swFeat, swLockFeat As SldWorks.Feature
Dim swFeatMgr As SldWorks.FeatureManager
```

Sub main()

```
    Set swApp = Application.SldWorks
    Set swModel = swApp.ActiveDoc
    Set swSelMgr = swModel.SelectionManager
    Set swFeatMgr = swModel.FeatureManager
    Set swFeat = swSelMgr.GetSelectedObject6(1, -1)
    Set swLockFeat = swFeatMgr.InsertFeatureLock
```

End Sub

Remarks

After calling this method, a temporary freeze bar displays after the feature in the FeatureManager design tree. If you move the freeze bar to the top, the feature unlocks and the freeze bar disappears. To permanently enable the freeze bar, select **Tools > Options > System Options > General > Enable Freeze bar**.

This method is analogous to freezing a feature. For more information, see **SOLIDWORKS Design user-interface help > Parts and Features > Features > Feature Freeze > Freezing Features**.

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[IFeature::IsFrozen Method \(\)](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertFeatureTreeFolder Method (IFeatureManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertFeatureTreeFolder Method (IFeatureManager)

Obsolete. Superseded by [IFeatureManager::InsertFeatureTreeFolder2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function InsertFeatureTreeFolder() As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager
Dim value As System.Boolean
```

```
value = instance.InsertFeatureTreeFolder()
```

C#

```
System.bool InsertFeatureTreeFolder()
```

C++/CLI

```
System.bool InsertFeatureTreeFolder();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [FeatureManager.InsertFeatureTreeFolder](#).

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

InsertFeatureTreeFolder2 Method (IFeatureManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertFeatureTreeFolder2 Method (IFeatureManager)

Inserts a folder in the FeatureManager design tree for pre-selected features or components.

.NET Syntax

Visual Basic (Declaration)

```
Function InsertFeatureTreeFolder2( _  
    ByVal Type As System.Integer _  
) As Feature
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager  
Dim Type As System.Integer  
Dim value As Feature
```

```
value = instance.InsertFeatureTreeFolder2(Type)
```

C#

```
Feature InsertFeatureTreeFolder2(  
    System.int Type  
)
```

C++/CLI

```
Feature^ InsertFeatureTreeFolder2(  
&    System.int Type  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Type

Type of folder to insert as defined by [swFeatureTreeFolderType_e](#)

Return Value

Newly created folder [feature](#)

Visual Basic for Applications (VBA) Syntax

See [FeatureManager.InsertFeatureTreeFolder2](#).

Example

[Move Assembly Components to New Folder \(C#\)](#)
[Move Assembly Components to New Folder \(VB.NET\)](#)
[Move Assembly Components to New Folder \(VBA\)](#)
[Specify IGES Levels and Values, Then Import IGES File \(C#\)](#)
[Specify IGES Levels and Values, Then Import IGES File \(VB.NET\)](#)
[Specify IGES Levels and Values, Then Import IGES File \(VBA\)](#)

Remarks

This method operates on the features that are selected before this method executes.

If creating...

A folder to contain preselected features

An empty folder

Then...

Certain types of features cannot be moved into that folder, such as the predefined planes that exist when a new part is created. These types of features are ignored by this method.

If multiple features are selected when this method runs, all of the features are moved into the new folder; however, these features must be consecutive. If they are not, only the first group of consecutive features are moved into the new folder.

The folder is created just before the first valid feature in the selections. The new folder cannot be placed before certain features, such as the predefined planes that exist when a new part is created.

Mold folders

When a surface feature is pre-selected, three sub-folders are created in the Surface Bodies folder: Cavity Surface Bodies, Core Surface Bodies, and Parting Surface Bodies

Because a folder cannot be created inside another folder, any selected features already inside a folder are ignored.

This method works with components and features.

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[IFeatureFolder Interface](#)

Availability

SOLIDWORKS 2003 SP2, Revision Number 11.2

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertFilletBeadFeature Method (IFeatureManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertFilletBeadFeature Method (IFeatureManager)

Obsolete. Superseded by [IFeatureManager.InsertFilletBeadFeature2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function InsertFilletBeadFeature( _  
    ByVal Type1 As System.Short, _  
    ByVal Size1 As System.Double, _  
    ByVal Length1 As System.Double, _  
    ByVal Pitch As System.Double, _  
    ByVal Type2 As System.Short, _  
    ByVal Size2 As System.Double, _  
    ByVal Length2 As System.Double, _  
    ByVal Flag As System.Integer, _  
    ByVal EdgeNum1 As System.Integer, _  
    ByVal DeSelEdge1 As System.Object, _  
    ByVal EdgeNum2 As System.Integer, _  
    ByVal DeSelEdge2 As System.Object _  
) As Feature
```

Visual Basic (Usage)

```
Dim instance As IFeatureManager  
Dim Type1 As System.Short  
Dim Size1 As System.Double  
Dim Length1 As System.Double  
Dim Pitch As System.Double  
Dim Type2 As System.Short  
Dim Size2 As System.Double  
Dim Length2 As System.Double
```



```

Dim Flag As System.Integer
Dim EdgeNum1 As System.Integer
Dim DeSelEdge1 As System.Object
Dim EdgeNum2 As System.Integer
Dim DeSelEdge2 As System.Object
Dim value As Feature

```

```

value = instance.InsertFilletBeadFeature(Type1, Size1, Length1, Pitch, Type2, Size2, Length2,
Flag, EdgeNum1, DeSelEdge1, EdgeNum2, DeSelEdge2)

```

C#

```

Feature InsertFilletBeadFeature(
    System.short Type1,
    System.double Size1,
    System.double Length1,
    System.double Pitch,
    System.short Type2,
    System.double Size2,
    System.double Length2,
    System.int Flag,
    System.int EdgeNum1,
    System.object DeSelEdge1,
    System.int EdgeNum2,
    System.object DeSelEdge2
)

```

C++/CLI

```

Feature^ InsertFilletBeadFeature(
    & System.short Type1,
    & System.double Size1,
    & System.double Length1,
    & System.double Pitch,
    & System.short Type2,
    & System.double Size2,
    & System.double Length2,
    & System.int Flag,
    & System.int EdgeNum1,
    & System.Object^ DeSelEdge1,
    & System.int EdgeNum2,
    & System.Object^ DeSelEdge2
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Type1

First side:

- 0 = Full length
- 1 = Intermittent
- 2 = Staggered

Size1

Size of fillet on first side

Length1

Length of fillet on first side

Pitch

Pitch, if Intermittent or Staggered on both sides

Type2

Second side:

- 0= Full length
- 1 = Intermittent
- 2= Staggered

Size2

Size of fillet on second side

Length2

Length of fillet on second side

Flag

- 0 = One-sided and no tangent propagation
- 1 = Two-sided
- 2 = Tangent propagation
- 3 = Two-sided and tangent propagation

EdgeNum1
Number of intersecting edges on first side
DeSelEdge1
Array indicating if intersecting edges are selected (0) or deselected (1) on first side

EdgeNum2
Number of intersecting edges on second side
DeSelEdge2
Array indicating if intersecting edges are selected (0) or deselected (1) on second side

Return Value

Pointer to the [IFeature](#) object

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::InsertFilletBeadFeature](#).

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[IWeldmentBeadFeatureData Interface](#)

Availability

SOLIDWORKS 2004 FCS, Revision Number 12.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

InsertFilletBeadFeature2 Method (IFeatureManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertFilletBeadFeature2 Method (IFeatureManager)

Inserts a fillet weld bead feature and also fills the gap between the pre-selected part bodies, if any.

Visual Basic Syntax

Visual Basic (Declaration)

```
Function InsertFilletBeadFeature2( _  
    ByVal Type1 As System.Short, _  
    ByVal Size1 As System.Double, _  
    ByVal Length1 As System.Double, _  
    ByVal PenetrationType1 As System.Short, _  
    ByVal PenetrationValue1 As System.Double, _  
    ByVal Pitch As System.Double, _  
    ByVal Type2 As System.Short, _  
    ByVal Size2 As System.Double, _  
    ByVal Length2 As System.Double, _  
    ByVal PenetrationType2 As System.Short, _  
    ByVal PenetrationValue2 As System.Double, _  
    ByVal Flag As System.Integer, _  
    ByVal EdgeNum1 As System.Integer, _  
    ByVal DeSelEdge1 As System.Object, _  
    ByVal EdgeNum2 As System.Integer, _  
    ByVal DeSelEdge2 As System.Object _  
) As Feature
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager  
Dim Type1 As System.Short  
Dim Size1 As System.Double  
Dim Length1 As System.Double  
Dim PenetrationType1 As System.Short  
Dim PenetrationValue1 As System.Double  
Dim Pitch As System.Double  
Dim Type2 As System.Short  
Dim Size2 As System.Double  
Dim Length2 As System.Double  
Dim PenetrationType2 As System.Short
```

```

Dim PenetrationValue2 As System.Double
Dim Flag As System.Integer
Dim EdgeNum1 As System.Integer
Dim DeSelEdge1 As System.Object
Dim EdgeNum2 As System.Integer
Dim DeSelEdge2 As System.Object
Dim value As Feature

```

```

value = instance.InsertFilletBeadFeature2(Type1, Size1, Length1, PenetrationType1,
PenetrationValue1, Pitch, Type2, Size2, Length2, PenetrationType2, PenetrationValue2, Flag,
EdgeNum1, DeSelEdge1, EdgeNum2, DeSelEdge2)

```

C#

```

Feature InsertFilletBeadFeature2(
    System.short Type1,
    System.double Size1,
    System.double Length1,
    System.short PenetrationType1,
    System.double PenetrationValue1,
    System.double Pitch,
    System.short Type2,
    System.double Size2,
    System.double Length2,
    System.short PenetrationType2,
    System.double PenetrationValue2,
    System.int Flag,
    System.int EdgeNum1,
    System.object DeSelEdge1,
    System.int EdgeNum2,
    System.object DeSelEdge2
)

```

C++/CLI

```

Feature^ InsertFilletBeadFeature2(
    & System.short Type1,
    & System.double Size1,
    & System.double Length1,
    & System.short PenetrationType1,
    & System.double PenetrationValue1,
    & System.double Pitch,
    & System.short Type2,
    & System.double Size2,
    & System.double Length2,
    & System.short PenetrationType2,
    & System.double PenetrationValue2,
    & System.int Flag,
    & System.int EdgeNum1,
    & System.Object^ DeSelEdge1,
    & System.int EdgeNum2,
    & System.Object^ DeSelEdge2
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Type1

First side:

- 0 = Full length
- 1 = Intermittent
- 2 = Staggered

Size1

Size of fillet on first side

Length1

Length of fillet on first side

PenetrationType1

First side:

- 0 = Full penetration
- 1 = Partial penetration

- 2 = No penetration

PenetrationValue1

If *penetrationType1* is set to 1 (Partial penetration), then set its value

Pitch

Pitch, if Intermittent or Staggered on both sides

Type2

Second side:

- 0= Full length
- 1 = Intermittent
- 2= Staggered

Size2

Size of fillet on second side

Length2

Length of fillet on second side

PenetrationType2

Second side:

- 0 = Full penetration
- 1 = Partial penetration
- 2 = No penetration

PenetrationValue2

If *penetrationType2* is set to 1 (Partial penetration), then set its value

Flag

- 0 = One-sided and no tangent propagation
- 1 = Two-sided
- 2 = Tangent propagation
- 3 = Two-sided and tangent propagation

EdgeNum1

Number of intersecting edges on first side

DeSelEdge1

Array indicating if intersecting edges are selected (0) or deselected (1) on first side

EdgeNum2

Number of intersecting edges on second side

DeSelEdge2

Array indicating if intersecting edges are selected (0) or deselected (1) on second side

Return Value

[Feature](#)

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::InsertFilletBeadFeature2](#).

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[IEndCapFeatureData Interface](#)
[IFeatureManager::InsertEndCapFeature Method](#)
[IFeatureManager::InsertStructuralWeldment2 Method](#)
[IFeatureManager::InsertSubWeldFolder Method](#)
[IFeatureManager::InsertWeldmentCutList Method](#)
[IFeatureManager::InsertWeldmentFeature Method](#)
[IFeatureManager::InsertWeldmentTrimFeature Method](#)
[IGussetFeatureData Interface](#)
[IStructuralMemberFeatureData Interface](#)
[IWeldmentBeadFeatureData Interface](#)
[IWeldmentCutListAnnotation Interface](#)
[IWeldmentCutListFeature Interface](#)
[IWeldBead Interface](#)
[IFeatureManager::InsertFilletBeadFeature3 Method](#)

Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertFilletBeadFeature3 Method (IFeatureManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertFilletBeadFeature3 Method (IFeatureManager)

Inserts fillet weld bead features for the specified face sets.

Visual Basic (Declaration)

```
Function InsertFilletBeadFeature3( _
    ByVal Type1 As System.Short, _
    ByVal Size1 As System.Double, _
    ByVal Length1 As System.Double, _
    ByVal PenetrationType1 As System.Short, _
    ByVal PenetrationValue1 As System.Double, _
    ByVal Pitch As System.Double, _
    ByVal Type2 As System.Short, _
    ByVal Size2 As System.Double, _
    ByVal Length2 As System.Double, _
    ByVal PenetrationType2 As System.Short, _
    ByVal PenetrationValue2 As System.Double, _
    ByVal Flag As System.Integer, _
    ByVal EdgeNum1 As System.Integer, _
    ByVal DeSelEdge1 As System.Object, _
    ByVal EdgeNum2 As System.Integer, _
    ByVal DeSelEdge2 As System.Object, _
    ByVal FaceSet1 As System.Object, _
    ByVal FaceSet2 As System.Object _
) As Feature
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager
Dim Type1 As System.Short
Dim Size1 As System.Double
Dim Length1 As System.Double
Dim PenetrationType1 As System.Short
Dim PenetrationValue1 As System.Double
Dim Pitch As System.Double
Dim Type2 As System.Short
Dim Size2 As System.Double
Dim Length2 As System.Double
Dim PenetrationType2 As System.Short
Dim PenetrationValue2 As System.Double
Dim Flag As System.Integer
Dim EdgeNum1 As System.Integer
Dim DeSelEdge1 As System.Object
Dim EdgeNum2 As System.Integer
Dim DeSelEdge2 As System.Object
Dim FaceSet1 As System.Object
Dim FaceSet2 As System.Object
Dim value As Feature

value = instance.InsertFilletBeadFeature3(Type1, Size1, Length1, PenetrationType1,
PenetrationValue1, Pitch, Type2, Size2, Length2, PenetrationType2, PenetrationValue2, Flag,
EdgeNum1, DeSelEdge1, EdgeNum2, DeSelEdge2, FaceSet1, FaceSet2)
```

C#

```
Feature InsertFilletBeadFeature3(
    System.short Type1,
    System.double Size1,
    System.double Length1,
    System.short PenetrationType1,
    System.double PenetrationValue1,
    System.double Pitch,
    System.short Type2,
    System.double Size2,
    System.double Length2,
    System.short PenetrationType2,
    System.double PenetrationValue2,
    System.int Flag,
    System.int EdgeNum1,
    System.object DeSelEdge1,
```

```

    System.int EdgeNum2,
    System.object DeSelEdge2,
    System.object FaceSet1,
    System.object FaceSet2
)

```

C++/CLI

```

Feature^ InsertFilletBeadFeature3(
& System.short Type1,
& System.double Size1,
& System.double Length1,
& System.short PenetrationType1,
& System.double PenetrationValue1,
& System.double Pitch,
& System.short Type2,
& System.double Size2,
& System.double Length2,
& System.short PenetrationType2,
& System.double PenetrationValue2,
& System.int Flag,
& System.int EdgeNum1,
& System.Object^ DeSelEdge1,
& System.int EdgeNum2,
& System.Object^ DeSelEdge2,
& System.Object^ FaceSet1,
& System.Object^ FaceSet2
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Type1

First side:

- 0 = Full length
- 1 = Intermittent
- 2 = Staggered

Size1

Size of fillet on first side

Length1

Length of fillet on first side

PenetrationType1

First side:

- 0 = Full penetration
- 1 = Partial penetration
- 2 = No penetration

PenetrationValue1

If penetrationType1 is set to 1 (Partial penetration), then set its value

Pitch

Pitch, if Intermittent or Staggered on both sides

Type2

Second side:

- 0= Full length
- 1 = Intermittent
- 2= Staggered

Size2

Size of fillet on second side

Length2

Length of fillet on second side

PenetrationType2

Second side:

- 0 = Full penetration
- 1 = Partial penetration
- 2 = No penetration

PenetrationValue2

If penetrationType2 is set to 1 (Partial penetration), then set its value

Flag

- 0 = One-sided and no tangent propagation

- 1 = Two-sided
- 2 = Tangent propagation
- 3 = Two-sided and tangent propagation

EdgeNum1

Number of intersecting edges on first side

DeSelEdge1

Array indicating if intersecting edges are selected (0) or deselected (1) on first side

EdgeNum2

Number of intersecting edges on second side

DeSelEdge2

Array indicating if intersecting edges are selected (0) or deselected (1) on second side

FaceSet1

Array of side 1 [faces](#) whose intersecting edges get fillet beads

FaceSet2

Array of side 2 [faces](#) whose intersecting edges get fillet beads

Return Value

[Feature](#)

Visual Basic for Applications (VBA) Syntax

See [FeatureManager.InsertFilletBeadFeature3](#).

Example

[Insert Weldment Features \(VBA\)](#)

[Insert Weldment Features \(VB.NET\)](#)

[Insert Weldment Features \(C#\)](#)

Remarks

This interface specifies two face arrays, one for each side of a part feature. The fillet bead is applied to the edges where each face intersects for each array specified.

See Also

[IFeatureManager Interface](#)

[IFeatureManager Members](#)

[WeldmentBeadFeatureData Interface](#)

[IFeatureManager.InsertFilletBeadFeature2 Method](#)

Availability

SOLIDWORKS 2010 FCS, Revision Number 18.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertFillSurface Method (IFeatureManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertFillSurface Method (IFeatureManager)

Obsolete. Superseded by [IFeatureManager.InsertFillSurface2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function InsertFillSurface( _  
    ByVal Resolution As System.Integer _  
) As Feature
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager  
Dim Resolution As System.Integer  
Dim value As Feature
```

```
value = instance.InsertFillSurface(Resolution)
```

C#

```
Feature InsertFillSurface(  
    System.int Resolution  
)
```

C++/CLI

```
Feature^ InsertFillSurface(  
&    System.int Resolution  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Resolution

Visual Basic for Applications (VBA) Syntax

See [FeatureManager.InsertFillSurface](#).

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

InsertFillSurface2 Method (IFeatureManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertFillSurface2 Method (IFeatureManager)

Inserts a fill-surface feature in the model.

.NET Syntax

Visual Basic (Declaration)

```
Function InsertFillSurface2( _  
    ByVal Resolutions As System.Integer, _  
    ByVal Options As System.Integer, _  
    ByVal VPatchBoundaries As System.Object, _  
    ByVal VCurvatureControlType As System.Object, _  
    ByVal VFaces As System.Object, _  
    ByVal VConstraintCurves As System.Object _  
) As Feature
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager  
Dim Resolutions As System.Integer  
Dim Options As System.Integer  
Dim VPatchBoundaries As System.Object  
Dim VCurvatureControlType As System.Object  
Dim VFaces As System.Object  
Dim VConstraintCurves As System.Object  
Dim value As Feature
```

```
value = instance.InsertFillSurface2(Resolutions, Options, VPatchBoundaries, VCurvatureControlType,  
VFaces, VConstraintCurves)
```

C#

```
Feature InsertFillSurface2(  
    System.int Resolutions,  
    System.int Options,  
    System.object VPatchBoundaries,  
    System.object VCurvatureControlType,  
    System.object VFaces,
```

```
System.Object VConstraintCurves  
)
```

C++/CLI

```
Feature^ InsertFillSurface2(  
& System.Int Resolutions,  
& System.Int Options,  
& System.Object^ VPatchBoundaries,  
& System.Object^ VCurvatureControlType,  
& System.Object^ VFaces,  
& System.Object^ VConstraintCurves  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Resolutions

Controls the resolution or quality of the surface (see **Remarks**)

Options

Options as defined in [swFeatureFillSurfaceOptions_e](#)

VPatchBoundaries

Array of [edges](#) or [sketches](#) for the patch boundaries

VCurvatureControlType

Array of curve control methods as defined in [swContactType_e](#)

VFaces

Array of [faces](#) to use for direction

VConstraintCurves

Array of constraint curves ([edges](#) or [sketches](#))

Return Value

Pointer to the [IFeature](#) object

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::InsertFillSurface2](#).

Example

[Insert Fill-surface Feature \(C#\)](#)
[Insert Fill-surface Feature \(VB.NET\)](#)
[Insert Fill-surface Feature \(VBA\)](#)
[Get and Fill Gaps in Body \(C#\)](#)
[Get and Fill Gaps in Body \(VB.NET\)](#)
[Get and Fill Gaps in Body \(VBA\)](#)

Remarks

You must use [IModelDocExtension::SelectByID2](#) and the following Mark values to select edges that bound the surface to be filled:

- Boundary curves = 1
- Boundary with contact curvature control = 257
- Boundary with tangent curvature control = 513
- Constraint curves or internal curves = 4

The resolution argument can be set to 1, 2, or 3. The higher the value, the better the resolution.

Use the [IBody2::Diagnose](#) and the [IDiagnoseResult](#) APIs to get the gaps to fill.

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[IFillSurfaceFeatureData Interface](#)

Availability

SOLIDWORKS 2007 FCS, Revision Number 15.0



Obsolete. Superseded by [IFeatureManager::InsertFlattenSurface2](#).

Visual Basic Syntax

Visual Basic (Declaration)

```
Function InsertFlattenSurface( _  
    ByVal AccuracyFactor As System.Integer _  
) As Feature
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager  
Dim AccuracyFactor As System.Integer  
Dim value As Feature  
  
value = instance.InsertFlattenSurface(AccuracyFactor)
```

C#

```
Feature InsertFlattenSurface(  
    System.int AccuracyFactor  
)
```

C++/CLI

```
Feature^ InsertFlattenSurface(  
& System.int AccuracyFactor  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

AccuracyFactor
10 >= Accuracy of flattened triangle mesh >= 1; 1 is highest accuracy

Return Value

Surface-flatten [feature](#)

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::InsertFlattenSurface](#).

Remarks

Before calling this method, select...	By calling IModelDocExtension::SelectByID2 with Mark...
Faces or surfaces to flatten	1
Anchor point from which to flatten	16
Edges to guide the shape and length of the flattened surface	2

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[ISurfaceFlattenFeatureData Interface](#)

Availability

SOLIDWORKS 2015 FCS, Revision Number 23.0



Inserts a surface-flatten feature in the model.

 .NET Syntax

```
Visual Basic (Declaration)

Function InsertFlattenSurface2( _
    ByVal AccuracyFactor As System.Integer, _
    ByVal ShouldUseTears As System.Boolean _
) As Feature
```

```
Visual Basic (Usage)   Copy Code

Dim instance As IFeatureManager
Dim AccuracyFactor As System.Integer
Dim ShouldUseTears As System.Boolean
Dim value As Feature

value = instance.InsertFlattenSurface2(AccuracyFactor, ShouldUseTears)
```

```
C#

Feature InsertFlattenSurface2(
    System.int AccuracyFactor,
    System.bool ShouldUseTears
)
```

```
C++/CLI

Feature^ InsertFlattenSurface2(
&    System.int AccuracyFactor,
&    System.bool ShouldUseTears
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

AccuracyFactor
10 >= Accuracy of flattened triangle mesh >= 1; 1 is highest accuracy
ShouldUseTears
True to use relief cuts, false to not

Return Value

Surface-flatten [feature](#)

 Visual Basic for Applications (VBA) Syntax

See [FeatureManager.InsertFlattenSurface2](#).

 Example

[Get Data for Surface-flatten Feature \(C#\)](#)
[Get Data for Surface-flatten Feature \(VB.NET\)](#)
[Get Data for Surface-flatten Feature \(VBA\)](#)

 Remarks

Before calling this method, select...	By calling IModelDocExtension::SelectByID2 with Mark...
Faces or surfaces to flatten	1
Edges to guide the shape and length of the flattened surface	2
Map edges	4
Tear edges	8
Anchor point from which to flatten	16

Map edges:

- are entities to map to the flattened surface. By default, any entity that lies on top of a face or surface to flatten is ignored unless the user selects the entity as a map edge.
- are useful when users want to see where a selected entity on the original body would lie on a flattened surface. For example, users might select edges as map lines to serve as bend lines in the flattened surface.
- can be any of these types of selected entities:
 - edges.
 - reference curves,
 - 2D/3D splines,

- sketches, and
- sketch text.

Tear edges:

- indicate where to make relief cuts before flattening the surface. During flattening, regions of the flattened surface are stretched while other regions are compressed. Adding tear edge through regions of high stress can reduce these deformations, which increases the accuracy of the flattened surface.
- allow users to flatten closed surfaces, such as spheres or entire solid bodies, that would otherwise not be flattened.
- can be any of these types of selected entities:
 - edges,
 - reference curves, and
 - 2D/3D splines.
- must each intersect the face or surface being flattened while touching one boundary edge, which is an edge adjacent to exactly one face or surface.
- are only applied when ShouldUseTears is true.

An anchor point can be a vertex or point on an edge.

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[ISurfaceFlattenFeatureData Interface](#)

Availability

SOLIDWORKS 2016 FCS, Revision Number 24.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertFlexFeature Method (IFeatureManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertFlexFeature Method (IFeatureManager)

Inserts a Flex feature using the selected solid or surface body.

.NET Syntax

Visual Basic (Declaration)

```
Function InsertFlexFeature( _
    ByVal RotX As System.Double, _
    ByVal RotY As System.Double, _
    ByVal RotZ As System.Double, _
    ByVal TanX As System.Double, _
    ByVal TanY As System.Double, _
    ByVal TanZ As System.Double, _
    ByVal RadX As System.Double, _
    ByVal RadY As System.Double, _
    ByVal RadZ As System.Double, _
    ByVal Angle As System.Double, _
    ByVal PivotX As System.Double, _
    ByVal PivotY As System.Double, _
    ByVal PivotZ As System.Double, _
    ByVal Type As System.Integer, _
    ByVal LeftTrim As System.Double, _
    ByVal RightTrim As System.Double, _
    ByVal HardEdges As System.Boolean _
) As Feature
```

Visual Basic (Usage)

```
Dim instance As IFeatureManager
Dim RotX As System.Double
Dim RotY As System.Double
Dim RotZ As System.Double
Dim TanX As System.Double
Dim TanY As System.Double
Dim TanZ As System.Double
Dim RadX As System.Double
Dim RadY As System.Double
Dim RadZ As System.Double
Dim Angle As System.Double
Dim PivotX As System.Double
Dim PivotY As System.Double
Dim PivotZ As System.Double
```



```

Dim Type As System.Integer
Dim LeftTrim As System.Double
Dim RightTrim As System.Double
Dim HardEdges As System.Boolean
Dim value As Feature

```

```

value = instance.InsertFlexFeature(RotX, RotY, RotZ, TanX, TanY, TanZ, RadX, RadY, RadZ, Angle,
PivotX, PivotY, PivotZ, Type, LeftTrim, RightTrim, HardEdges)

```

C#

```

Feature InsertFlexFeature(
    System.double RotX,
    System.double RotY,
    System.double RotZ,
    System.double TanX,
    System.double TanY,
    System.double TanZ,
    System.double RadX,
    System.double RadY,
    System.double RadZ,
    System.double Angle,
    System.double PivotX,
    System.double PivotY,
    System.double PivotZ,
    System.int Type,
    System.double LeftTrim,
    System.double RightTrim,
    System.bool HardEdges
)

```

C++/CLI

```

Feature^ InsertFlexFeature(
    & System.double RotX,
    & System.double RotY,
    & System.double RotZ,
    & System.double TanX,
    & System.double TanY,
    & System.double TanZ,
    & System.double RadX,
    & System.double RadY,
    & System.double RadZ,
    & System.double Angle,
    & System.double PivotX,
    & System.double PivotY,
    & System.double PivotZ,
    & System.int Type,
    & System.double LeftTrim,
    & System.double RightTrim,
    & System.bool HardEdges
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

RotX
x coordinate for the rotation axis

RotY
y coordinate for the rotation axis

RotZ
z coordinate for the rotation axis

TanX
x coordinate for the tangent axis

TanY
y coordinate for the tangent axis

TanZ
z coordinate for the tangent axis

RadX
x coordinate for the radius axis

RadY
y coordinate for the radius axis

RadZ
z coordinate for the radius axis

Angle

Angle

PivotX

x coordinate for the pivot triad reference

PivotY

y coordinate for the pivot triad reference

PivotZ

z coordinate for the pivot triad reference

Type

- -1 =None

- 0 = Bending
- 1 = Twisting
- 2 = Tapering
- 3 = Stretching
- 4 = Experimental

LeftTrim

Left trimming distance

RightTrim

Right trimming distance

HardEdges

True to embed edges where the trim planes intersect the body, false to create smoother continuous faces

Return Value

Pointer to the [IFeature](#) object

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::InsertFlexFeature](#).

See Also

[IFeatureManager Interface](#)

[IFeatureManager Members](#)

Availability

SOLIDWORKS 2005 FCS, Revision Number 13.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertFormToolFeature Method (IFeatureManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertFormToolFeature Method (IFeatureManager)

Obsolete. Superseded by [IFeatureManager::CreateDefinition](#) and [IFeatureManager::CreateFeature](#).

.NET Syntax

Visual Basic (Declaration)

```
Function InsertFormToolFeature( _  
    ByVal Path As System.String, _  
    ByVal Flip As System.Boolean, _  
    ByVal RotAngle As System.Double, _  
    ByVal Config As System.String, _  
    ByVal OverrideDoc As System.Boolean, _  
    ByVal ShowPunch As System.Boolean, _  
    ByVal ShowProfile As System.Boolean, _  
    ByVal ShowCenter As System.Boolean, _  
    ByVal LinkToPart As System.Boolean _  
) As Feature
```

Visual Basic (Usage)

```
Dim instance As IFeatureManager  
Dim Path As System.String  
Dim Flip As System.Boolean  
Dim RotAngle As System.Double  
Dim Config As System.String
```



```

Dim OverrideDoc As System.Boolean
Dim ShowPunch As System.Boolean
Dim ShowProfile As System.Boolean
Dim ShowCenter As System.Boolean
Dim LinkToPart As System.Boolean
Dim value As Feature

```

```

value = instance.InsertFormToolFeature(Path, Flip, RotAngle, Config, OverrideDoc, ShowPunch,
ShowProfile, ShowCenter, LinkToPart)

```

C#

```

Feature InsertFormToolFeature(
    System.string Path,
    System.bool Flip,
    System.double RotAngle,
    System.string Config,
    System.bool OverrideDoc,
    System.bool ShowPunch,
    System.bool ShowProfile,
    System.bool ShowCenter,
    System.bool LinkToPart
)

```

C++/CLI

```

Feature^ InsertFormToolFeature(
& System.String^ Path,
& System.bool Flip,
& System.double RotAngle,
& System.String^ Config,
& System.bool OverrideDoc,
& System.bool ShowPunch,
& System.bool ShowProfile,
& System.bool ShowCenter,
& System.bool LinkToPart
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Path

Pathname of the forming tool part file in the Design Library

Flip

Whether to reverse the direction of the cut of the forming tool when inserted

RotAngle

Angle of the forming tool

Config

Name of the configuration of the forming tool to insert

OverrideDoc

True to override the document settings in **Tools > Options > Document Properties > Sheet Metal**, false to not

ShowPunch

True to display the cut of the forming tool when the part is flattened, false to not; valid only if OverrideDoc = true

ShowProfile

True to display the placement sketch of the forming tool when the part is flattened, false to not; valid only if OverrideDoc = true

ShowCenter

True to display the center mark where the forming tool is located in the flat pattern, false to not; valid only if OverrideDoc = true

LinkToPart

True to link the forming tool feature to its part in the Design Library, false to not

Return Value

[IFeature](#)

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::InsertFormToolFeature](#).

Remarks

Before calling this method, select either a face or a 2D sketch containing points. If you select:

- face, a single instance of the Design Library forming tool is placed on it.
- sketch containing points, an instance of the Design Library forming tool is placed at each point in the sketch.

To create your own forming tool, call [IFeatureManager::CreateFormTool](#).

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[View.InsertPunchTable Method](#)

Availability

SOLIDWORKS 2012 FCS, Revision Number 20.0